

Multimodal Fusion Strategies for Pulmonary Embolism Diagnosis and Prognosis Using Heterogeneous CT Architectures

Luke Hidveghy

Supervisor: Debmalya Mandal

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Department of Computer Science

University of Warwick

Abstract

Pulmonary embolism (PE) remains a challenging task for both diagnosis and prognosis because clinically relevant information is distributed across imaging and structured clinical data. This dissertation investigates how the stage at which electronic health record (EHR) data are integrated affects performance in multimodal PE prediction when combined with a heterogeneous computed tomography (CT) ensemble. Using the INSPECT dataset, models were developed for both PE diagnosis and 12-month mortality prediction. Imaging models comprised heterogeneous 2D, 2.5D, and 3D CT architectures, while the EHR branch used structured clinical features with Logistic Regression and XGBoost. These components were combined using two alternative multimodal strategies. In flat fusion models, CT and EHR predictions were merged at the same stage. In hierarchical fusion, CT predictions were first stacked and then combined with EHR. Results showed that EHR-only models provided strong standalone performance and outperformed individual CT-only models on both tasks, while CT-only stacking produced only modest gains. For diagnosis, the strongest flat and hierarchical fusion models performed almost identically (Area Under the Precision-Recall Curve (AUPRC) 0.7288 vs 0.7266), suggesting that the main gain arose from combining modalities rather than from the stage of fusion. For prognosis, hierarchical fusion outperformed the best flat model (AUPRC 0.4995 vs 0.4707; Area Under the Receiver Operating Characteristic curve (AUROC) 0.8622 vs 0.8349), indicating that fusion timing affected performance within the multimodal setting. However, the best overall prognosis performance was achieved by the EHR-only XGB baseline, suggesting that CT provided limited incremental value beyond structured clinical features in this implementation. Overall, the findings suggest that the value of complex fusion architecture is task dependent with results showing it is limited for diagnosis, but more consequential for longer-horizon prognosis within the multimodal setting.

Keywords

Pulmonary embolism; multimodal fusion; computed tomography pulmonary angiography; electronic health records; stacked ensemble; diagnosis; prognosis

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Chapter 1

Introduction

1.1 Background

This dissertation investigates whether the stage at which electronic health record (EHR) data are integrated affects multimodal performance in Pulmonary Embolism (PE) diagnosis and 12-month mortality prognosis when combined with a heterogeneous CT ensemble. Pulmonary embolism is a condition in which a blood clot travels to the lungs and blocks an artery. These clots often originate in the deep veins of the leg, a condition known as deep vein thrombosis (DVT). Pulmonary embolism can cause a range of symptoms such as shortness of breath, chest pain, and a rapid heart rate. The main imaging test used to diagnose it is computed tomography pulmonary angiography (CTPA), which allows doctors to examine the blood vessels in the lungs for signs of blockage. Blood tests may also be used, particularly to measure D-dimer, a substance released when blood clots are broken down in the body [1].

Despite the availability of established tests, both the diagnosis and prognosis of pulmonary embolism remain challenging in clinical practice. Small clots can be difficult to detect, particularly when CTPA image quality is affected by respiratory motion and noise, contributing to variability in radiologist reporting [2]. Diagnostic assessment is therefore often supported by clinical probability scores such as the Wells and Geneva scores [3]. Following diagnosis, prognostic stratification is used to estimate short-term mortality risk, commonly through the Pulmonary Embolism Severity Index or simplified PESI [4]. However, these approaches rely on a limited set of clinical features and may not fully capture the complexity of the available imaging and clinical data. This creates scope for machine learning methods that can integrate a larger range of information to support both diagnosis and prognosis, and improve robustness to noisy imaging data.

Machine learning approaches for pulmonary embolism diagnosis and prognosis have used multimodal fusion to combine imaging with clinical data. This reflects the range of information considered in clinical practice [5, 6, 7]. In addition, ensemble methods have been

applied to the diagnosis of pulmonary embolism from CT scans to address the limitations of individual architectures [8, 9]. However, it remains unclear whether structured clinical information is most useful when integrated with individual CT models before ensemble stacking, or when introduced only after heterogeneous CT predictions have been stacked.

1.2 Research Aim and Objectives

This dissertation aims to investigate the interaction between EHR data and a heterogeneous CT ensemble for pulmonary embolism diagnosis and 12-month mortality prognosis. Specifically, it compares integrating EHR as a base model within a single stacking layer with integrating EHR after the CT models have been stacked to assess whether additional fusion complexity yields meaningful performance gains. This comparison is motivated by the possibility that earlier integration may provide clinical context when aggregating individual imaging models. Alternatively, later integration may be more effective if EHR data are best combined with an already aggregated summary of heterogeneous CT predictions. This work does not aim to propose new fusion strategies or ensemble architectures, but rather to evaluate how the stage of EHR integration affects multimodal performance.

The objectives of this dissertation are as follows:

1. To evaluate the performance of heterogeneous CT ensembles for pulmonary embolism diagnosis and prognosis.
2. To assess the effect of incorporating EHR data before CT stacking.
3. To assess the effect of incorporating EHR data after CT stacking.
4. To compare these integration strategies using discrimination metrics, clinically relevant operating points, and subgroup analysis.
5. To analyse the contribution of imaging and clinical information to overall model performance through SHapley Additive exPlanations (SHAP) values [10].

Chapter 2

Literature Review

2.1 Pulmonary Embolism as a Diagnostic and Prognostic Challenge

Pulmonary embolism is the third leading cause of cardiovascular death [11], while cardiovascular diseases more broadly are the leading cause of death worldwide [12]. PE is therefore an important condition in which timely diagnosis and appropriate risk stratification are essential. Despite advances in diagnosis and treatment, important challenges remain in both detecting PE and identifying patients at increased risk of adverse outcomes. Clinically relevant information is distributed across pre-test assessment, laboratory testing, and imaging, meaning that both diagnosis and prognosis depend on integrating multiple imperfect sources of evidence [13].

Diagnosis of pulmonary embolism often uses a multimodal clinical pathway. This starts with assessment of pre-test probability using clinical prediction rules such as the Wells or Geneva scores [13]. Although these scores are widely validated and clinically useful for estimating pre-test probability, they are primarily designed as bedside decision tools rather than as fully individualised diagnostic models [14]. In patients with low or intermediate clinical probability, D-dimer testing is commonly used, and a negative result can help rule out PE because of its high sensitivity [15]. However, the test lacks specificity and cannot confirm PE on its own [16]. Definitive assessment, therefore, usually requires computed tomography pulmonary angiography (CTPA), a contrast-enhanced CT scan that visualises the pulmonary arteries and allows emboli to be identified as filling defects. Although CTPA is central to PE diagnosis, detection of small or subtle emboli remains difficult, and image interpretation may be affected by motion artefacts, suboptimal contrast opacification, and inter-reader variability among radiologists [2].

Beyond diagnosis, prognostic stratification is also central to PE management because outcomes vary substantially between patients. In current clinical practice, tools such

as the Pulmonary Embolism Severity Index and simplified PESI are commonly used to estimate short-term mortality risk and support early management decisions [4]. However, longer-term prognosis is likely to depend on a wider range of imaging and clinical factors than can be captured by compact bedside scores alone.

2.2 CT-Based Models for PE Diagnosis and Prognosis

A key early driver of CT-only pulmonary embolism modelling was the release of the RSNA Pulmonary Embolism CT Dataset, which provided a large multicentre benchmark for CT-based PE diagnosis [17]. Using this benchmark, PE diagnosis has been approached at multiple different levels, from slice- and scan-level classification to clot localisation, reflecting the fact that the task can be formulated in different ways depending on how emboli are represented in CT data. Representative work includes end-to-end volumetric CNNs such as PENet, which reported AUROC values of 0.84 on an internal hold-out test set and 0.85 on an external dataset [18]. However, later comparative work showed that CT-based PE detection performance depends strongly on modelling choices indicating that CT performance is sensitive to how the task is represented and optimised rather than being fixed by a single dominant architecture [19].

A recurring challenge for CT-only models is that emboli occupy only a small fraction of the volumetric scan, so individual slices vary considerably in their relevance to PE detection. Prior work has therefore aimed to restrict analysis to the most informative parts of the scan rather than treating the full volume as uniformly informative [20, 21]. For example, Tajbakhsh et al. proposed a vessel-oriented image representation intended to provide a more compact representation of embolus-related vascular structures [20]. More broadly, methods based on localisation, segmentation, or other intermediate focusing steps address the issue of PE-related signal being obscured when learning is performed over the full scan without further guidance [21, 22].

Compared with diagnosis, CT-based prognostic modelling in PE has been more limited and has frequently relied on predefined imaging biomarkers rather than fully end-to-end outcome prediction. Nevertheless, prior work has shown that routine CTPA contains clinically relevant prognostic information, particularly through imaging signs of right ventricular strain and clot burden that are associated with short-term adverse outcomes and early mortality [23, 24]. More recent radiomics-based studies likewise suggest that quantitative information extracted from CTPA may support PE severity assessment and mortality prediction, although this evidence base remains relatively limited and is still focused mainly on short-term outcomes rather than longer horizons [25]. This literature

suggests that a prognostic signal is present within CT, but that this has focused on hand-engineered or predefined features rather than on scan-level models that learn directly from routine image data.

2.3 Structured Clinical and EHR-Based Models for PE Diagnosis and Prognosis

In contrast to CT-based approaches, EHR-only and clinical-data models for pulmonary embolism estimate risk based on patient clinical context rather than from direct detection of emboli. In practice, for diagnosis, these approaches have traditionally taken the form of clinical decision rules, with a limited number of hand-selected variables [13]. More recent work has moved beyond simple score-based rules toward individualised structured-data prediction models. For example, Van Es et al. developed a clinical prediction model that used patient information and a D-dimer result to estimate an individual patient’s probability of having PE. They argue that traditional scoring rules were designed for simplicity in clinical use rather than for more personalised risk estimation [14]. This suggests that structured clinical variables contain meaningful diagnostic signal beyond rule-based scores, even though definitive diagnosis still depends on imaging.

Machine-learning studies have extended this structured-data paradigm by using larger longitudinal EHR feature sets. Banerjee et al. developed the Pulmonary Embolism Result Forecast Model (PERFORM), which used electronic medical records (EMR) to predict PE in patients referred for CT imaging. Their best model outperformed traditional rules such as Wells, PERC, and revised Geneva, suggesting that longitudinal EHR data may improve risk estimation compared to hand-crafted scores alone [26]. However, this work was framed primarily as decision support for CT triage. One challenge in structured EHR modelling is that missing and variably recorded data can introduce bias and affect cohort representativeness which can influence both model development and evaluation [27].

For prognosis, structured-data models are more established, but the literature has been dominated by short-term risk stratification tools rather than longer-horizon outcome prediction. The Pulmonary Embolism Severity Index (PESI) and simplified PESI (sPESI) use routinely available clinical variables to estimate mortality risk and are primarily used to identify patients at low short-term risk [28, 29]. Although some later studies suggest that these tools retain prognostic value beyond 30 days, longer-term structured-data prognostic modelling remains less mature. More recent structured-data-only studies have begun to explore this space, but they remain fewer in number and are still largely centred on short- or intermediate-term outcomes [30, 31].

Overall, the structured clinical data literature suggests that structured clinical data con-

tain substantial signal for both diagnosis and prognosis, but that most existing work either simplifies prediction into compact bedside scores or focuses on short-term outcomes.

2.4 Multimodal CT–EHR pipelines

Multimodal CT–EHR pipelines aim to combine the information from CTPA with structured clinical context. In pulmonary embolism, early work focused mainly on diagnosis. Huang et al. developed and compared multiple multimodal fusion models that combined CT imaging with EMR data for PE detection, using a cohort of 1,837 studies from 1,794 patients, and reported that multimodal models outperformed both imaging-only and EMR-only baselines. Zhi et al. likewise proposed a multimodal diagnostic framework that combined CT images with EHR data and reported better performance than either modality alone in a smaller cohort of 600 patients [6, 32]. However, these early studies were based on relatively small cohorts, which limited confidence in how robust their findings would be across different patient populations and institutions.

Subsequent work broadened the multimodal PE literature beyond small diagnostic case studies and towards more systematic benchmarking. RadFusion provided a paired CT–EHR benchmark for PE detection, enabling multimodal models to be compared not only in terms of predictive performance but also fairness across protected subgroups [5]. Multimodal prognostic modelling has also been developed. Cahan et al. developed fusion models for PE mortality prediction using CTPA together with tabular clinical data, showing that combined models could outperform single-modality approaches for risk stratification [33]. More recently, INSPECT introduced a substantially larger multimodal benchmark with linked structured EHR data and labels for both diagnostic and prognostic tasks, helping to make multimodal PE modelling more reproducible and comparable across settings [7].

To summarise, the existing multimodal CT–EHR literature shows that combining imaging with structured clinical data can improve both PE diagnosis and risk stratification compared with single-modality models, and that dedicated benchmark datasets have begun to make these more reproducible and comparable across studies [5, 6, 7].

2.5 Fusion and Stacking approaches in PE and Medical Imaging

Fusion methods are intended to exploit complementary information across modalities, but they differ in the stage at which the combination occurs. In multimodal biomedical learning, fusion strategies are commonly grouped into early, intermediate, and late fusion,

depending on whether modalities are combined at the input, feature, or decision level [34]. Prior reviews note that no single fusion strategy is optimal, as performance depends on the task [34]. In pulmonary embolism, multimodal studies have explored these choices. Huang et al. compared multiple CT-EMR fusion architectures for PE detection and found that a late-fusion model performed best among those tested [6]. In contrast, for PE mortality prediction, Cahan et al. evaluated early, late, and intermediate fusion strategies. They reported their strongest results with an intermediate-fusion model using bilinear attention and TabNet [33]. This literature indicates that the effectiveness of multimodal fusion in PE is task- and dataset-dependent, without identifying a single consistently superior strategy.

Stacking and ensemble methods are used to improve predictive performance by combining models with different strengths rather than relying on a single architecture. In CT-based medical imaging, stacked ensembles have been shown to outperform individual base models in tasks such as pancreas CT classification and CT radiomics classification of brain metastases. This supports the view that aggregating diverse learners can improve robustness and discrimination in clinical settings [35, 36]. In pulmonary embolism, recent ensemble-based studies report improved performance and robustness over single-model baselines on the RSNA benchmark [9]. Multimodal CT-EHR modelling is already well established in PE, whereas CT stacking has been studied less extensively. As stacking begins to emerge in this setting, it becomes increasingly worthwhile to examine how ensemble formation interacts with multimodal integration.

2.6 Research gap

Ensemble methods have been used in medical imaging to improve robustness and exploit complementary strengths across diverse models, but their interaction with multimodal fusion remains insufficiently studied in PE. Late fusion is closely related to ensemble learning because it combines the outputs of separate unimodal models rather than merging raw inputs or learned features. It remains unclear whether EHR information is most useful when introduced before CT ensemble formation, by providing clinical context to individual CT predictors, or after CT stacking, by complementing an already aggregated CT representation. This question is especially relevant when the CT component itself consists of heterogeneous 2D, 2.5D, and 3D architectures, which may respond differently to structured clinical information. In addition, much of the existing prognostic literature remains centred on short-term outcomes, leaving less evidence on how multimodal fusion and ensemble design affect longer-horizon PE prognosis. This dissertation therefore addresses a specific gap in the literature by evaluating how the timing of EHR integration interacts with heterogeneous CT ensemble stacking for both PE diagnosis and prognosis.

Chapter 3

Methodology

3.1 Study Dataset

This study used the Integrating Numerous Sources for Prognostic Evaluation of Clinical Timelines (INSPECT) dataset [7], a multimodal dataset for pulmonary embolism (PE) diagnosis and prognosis. INSPECT contains CT images, radiology report impression sections, and structured electronic health record (EHR) data, with labels for PE diagnosis, readmission, and mortality over 12 months. In the INSPECT benchmark, diagnostic PE labels were generated from radiology report impression sections using a trained NLP model described by Huang et al. In the present study, the CT and structured EHR modalities were used as model inputs.

In this study, prognostic evaluation focuses on 12-month mortality. Mortality was selected as the primary prognostic endpoint because it represents a relevant, long-term patient outcome. In addition, 12-month mortality allows assessment of whether multimodal machine learning can capture prognostic information over a longer time horizon than established early-risk stratification tools such as PESI and sPESI. It also avoids the limitations associated with the comparatively sparse 1-month and 6-month mortality labels.

3.2 Cohort Selection

The cohort was constructed by applying eligibility criteria across the study, series, and label metadata before combining these into a single cohort. Examinations were first restricted to PE/CTPA-related studies using regular-expression matching on the study description to identify terms consistent with pulmonary embolism CTPA, as described in Appendix A.1. The series were then filtered to retain only native diagnostic axial CT acquisitions, while reformatted, derived, non-diagnostic, sharp-kernel, bone-kernel, and lung-kernel series were excluded. These steps restricted analysis to the primary

image reconstruction relevant to PE assessment, excluding reconstructions optimised for structures other than the vascular soft-tissue appearance of interest to improve input consistency across scans.

Additional quality-control filtering was then applied. For each `image_id`, the series with the greatest number of slices was retained. This was used as a heuristic for selecting the most complete acquisition. Furthermore, series with fewer than 50 slices or slice thickness outside 0.5 to 3.0 mm were excluded as potentially incomplete or non-standard acquisitions. In parallel, only rows with complete diagnosis and mortality labels were retained to ensure target validity. The filtered imaging and label metadata were then combined using inner joins. This excluded scans not present across all required tables and reduced the cohort to 5,356 scans after merging.

After merging, only one eligible scan per patient was retained in order to prevent patient-level leakage across data splits and to reduce within-patient correlation. Where multiple eligible scans were available, the earliest scan was selected. This yielded 4,980 scans. For the diagnosis task, a 2:1 ratio of PE-negative to PE-positive cases was then enforced by random downsampling of the negative class, resulting in 4,665 scans. Due to storage and computational constraints, not all scans in this cohort could be downloaded and processed. The eligible scans were therefore partitioned into download batches based on the image identifier, and CT download and preprocessing were performed sequentially over these batches until resource limits were reached, giving 3,844 scans. A multimodal eligibility manifest was then constructed by checking each scan. Only scans with all four required inputs, namely 2D, 2.5D, 3D and the structured EHR feature set, were retained as fully multimodal. This required the removal of some scans with no corresponding data in the final EHR feature matrix. The final fully multimodal cohort comprised 3,443 scans, of which 1,104 were PE-positive and 2,339 were PE-negative. Although this final cohort should not be regarded as a random sample of all otherwise eligible scans, it retained most of the successfully processed scans.

To account for variation in structured EHR data density across patients, each patient was assigned to one of five EHR-density quintiles based on the number of non-zero structured EHR features. These quintiles were used together with the class label to define a composite stratification key for the patient-level training-test split. An 80/20 split yielded 2,754 training scans and 689 test scans for the diagnosis task. For the prognosis task, only PE-positive patients were included. This was because mortality prediction was intended to assess outcomes among patients with confirmed pulmonary embolism, for whom prognostic stratification is clinically meaningful. This yielded 883 training scans and 221 test scans, with 765 negative and 118 positive cases of 12-month mortality in the training set, and 192 negative and 29 positive cases in the test set. These train-test splits remained constant for all evaluation steps.

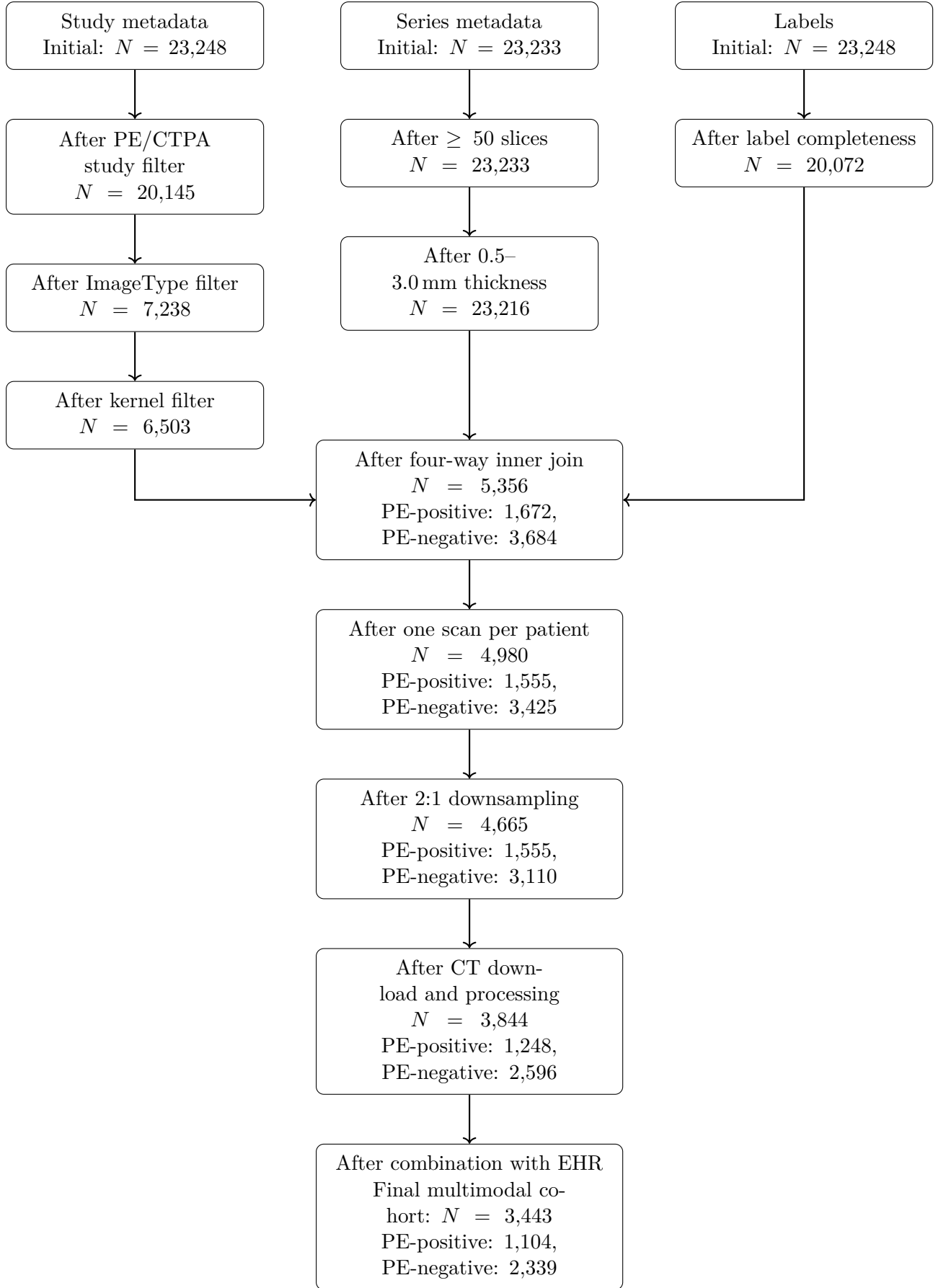


Figure 3.1: Flow diagram of cohort construction across study, series, and label metadata. PE-positive and PE-negative counts are available after merging the imaging and label streams.

3.3 Base Model Architectures

3.3.1 CT

Due to prior work suggesting that heterogeneous CT architectures can provide complementary information, the imaging ensemble comprised 2D, 2.5D, and 3D models. These were selected to capture different representations of the CT input. The 2D model was computationally efficient but had limited through-plane context. In contrast, the patch-based 3D model captured 3D spatial information but at a higher computational cost. The 2.5D model provided an intermediate representation by incorporating neighbouring slices while remaining more efficient than 3D modelling. Using all three, therefore, allowed the ensemble to combine complementary strengths across architectures rather than relying on a single representation of the scan.

CT Preprocessing

All CT pipelines first reoriented the scan, applied intensity scaling, and resampled the volume to a target spacing of $(1.0 \times 1.0 \times 2.0)$ mm in order to reduce variation arising from scanner and acquisition differences. A coarse per-slice mask was then constructed from voxels above -700 HU and collapsed across the candidate slices. This was then reduced to its largest connected component, and expanded by 8 mm. This defined a consistent body-centred crop while reducing background air. Where the resulting crop was too small, the full scan was retained as a fallback. Additional model-specific slice or patch selection was then applied as described below.

2D

The 2D model used a **ResNet-34** [37] backbone initialised with **IMAGENET1K_V1** weights. Slice-level features were aggregated into a scan-level representation using attention pooling implemented as a two-layer MLP with **Tanh** activation and a softmax over slices. A dropout rate of 0.3 was applied before classification. During training, all backbone layers except **layer4** were frozen, and separate learning rates were used for the backbone and classification head.

For 2D input generation, candidate slices were initially restricted to the central 70% of the resampled z-axis. Informative slice selection was then performed using a heuristic score based on body fraction, lung-band preference, central soft tissue, central vessel density, edge gradient, and position prior. The full definition of this score is provided in Appendix A.1. Forty-eight slices were selected per scan. Each selected slice was converted to a three-channel input using soft-tissue (50/350 HU), vascular (250/600 HU), and lung (-600/1500 HU) windows, and resized to 224×224 pixels.

2.5D

The 2.5D model represented each input as a stack of five neighbouring slices centred on an informative slice. To support this input format, the first convolutional layer of **ResNet-34** was modified to accept five input channels, with the additional weights initialised using the mean of the original RGB filters. A dropout rate of 0.4 was applied before classification. During training, all layers were frozen except **conv1**, **bn1**, and **layer4**.

Preprocessing followed the same pipeline as the 2D model. Forty-eight centre slices were selected per scan using the same heuristic scoring procedure. Edge slices were excluded as candidate centres to ensure that each input had a complete five-slice neighbourhood. A single vascular (250/600 HU) window was applied, and each five-slice stack was resized to 224×224 pixels per slice.

3D

The 3D model used MONAI’s **DenseNet121** [38, 39] backbone with a single input channel. A single vascular window centred at 250 HU with width 600 HU was used to emphasise contrast-enhanced vascular structures relevant to PE assessment. Rather than processing the full scan directly, the model operated on volumetric patches in order to reduce memory requirements while retaining local 3D spatial context. For each scan, four patches of size $160 \times 160 \times 64$ voxels were extracted from the resampled volume at a target spacing of $1.0 \times 1.0 \times 2.0$ mm. Patch placement was constrained by the same coarse body-region mask used in the 2D and 2.5D pipelines. Patch z-centres were determined using the informative-slice heuristic, while x- and y-centres were chosen deterministically from the given body-region and then paired with z-coordinates. Where the resampled scan was too shallow along the z-axis, edge-value padding was applied to ensure sufficient valid patch centres. Scan-level prediction was obtained by applying learned attention over patches using a linear layer followed by a softmax across patches. A dropout rate of 0.3 was applied before classification.

3.3.2 EHR

The INSPECT dataset provides structured health data for patients with CT scans. These data are stored across 100 parquet files and contain 225,029,469 coded events drawn from multiple medical vocabularies. The first step in preprocessing was to standardise these codes. Deprecated or invalid concepts were replaced with their updated equivalents. In addition, all codes were mapped to the corresponding standard concepts in the LOINC, RxNorm, and SNOMED vocabularies using the Observational Medical Outcomes Partnership (OMOP) vocabulary framework. This was done to improve consistency across sources and to reduce variation arising from different coding systems. Other adminis-

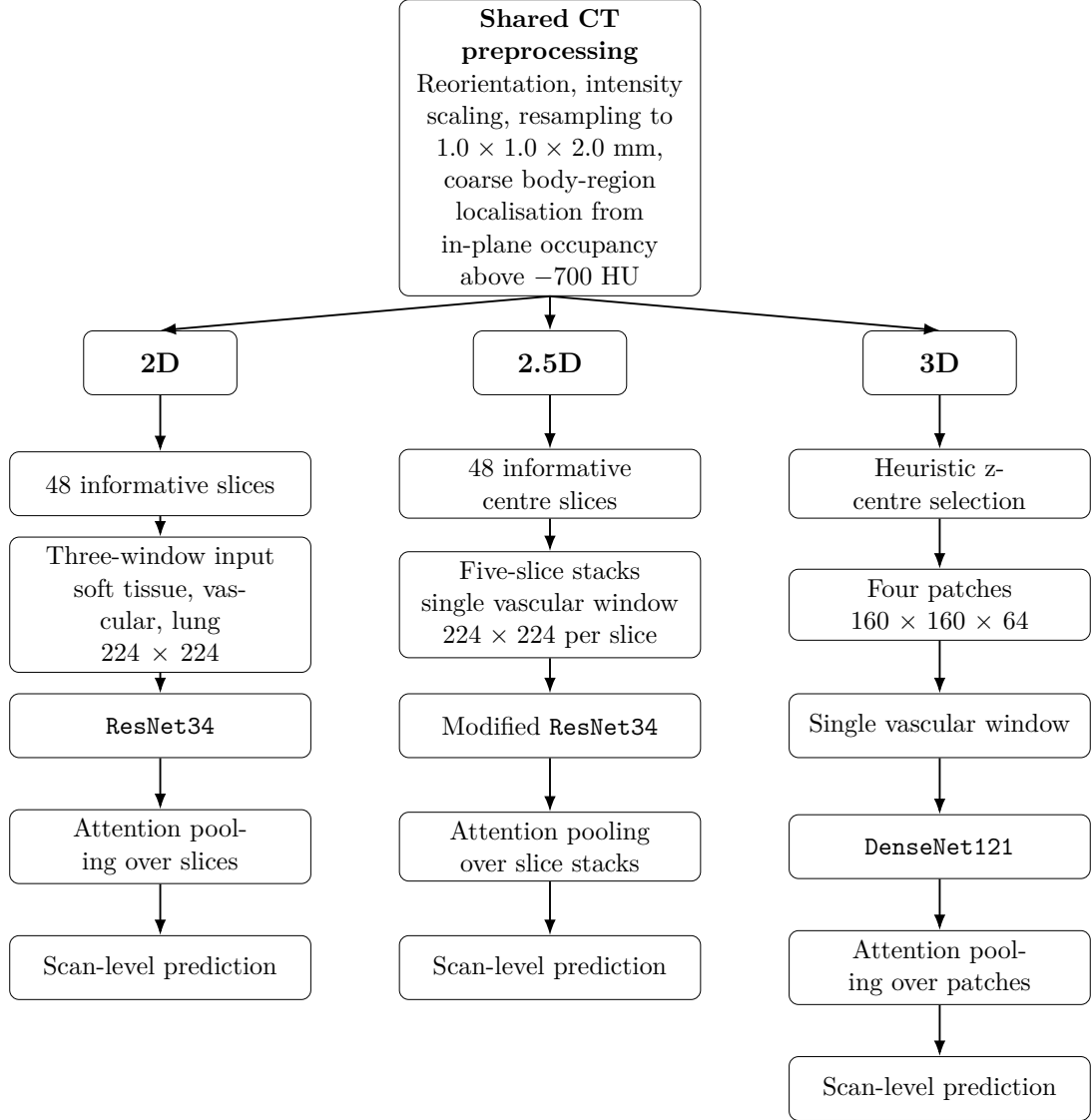


Figure 3.2: Overview of the three CT pipelines used in the heterogeneous imaging ensemble. All pipelines shared reorientation, intensity scaling, resampling, and coarse body-region localisation. The 2D pipeline operated on informative slices with three-window input channels, the 2.5D pipeline operated on five-slice stacks centred on informative slices, and the 3D pipeline operated on four volumetric patches with patch-level attention aggregation.

trative, demographic and workflow-related codes (for example `STANFORD_VISIT`, `Visit`, `Race`, `Ethnicity`, `Gender`, `CMS Place of Service`, `Medicare Specialty`, `NUCC`), were removed because, although they may reflect aspects of clinical decision-making, they were not directly clinically informative for the modelling task. In the case of CPT-4, this was removed due to additional licensing needs. Only events associated with the filtered set of patients retained in the CT selection pipeline were included. In addition, only events recorded at least 24 hours prior to the scan were used in order to prevent leakage from information generated after the imaging event and to preserve a clinically realistic prediction setting. Following code standardisation, further reduction was required for dimensionality and sparsity reasons, as the raw coded event space remained too large and heterogeneous for direct modelling. Vocabulary-specific feature grouping was therefore applied to produce a more compact and interpretable representation of the EHR data.

SNOMED events were rolled up using the OMOP `concept_ancestor` mapping. This maps each concept to a higher-level clinically meaningful condition-domain concept. One ancestor was assigned per concept, between two and four levels above the original concept. This was done to reduce sparsity by grouping identical or closely related conditions into a more compact representation while retaining clinically relevant structure. RxNorm codes were also rolled up using `concept_ancestor`. In this case, medication concepts were grouped to the ingredient level, using ancestors between levels 1 and 6. Multiple valid grouped ancestors were retained for the same drug concept. This differs from the SNOMED roll-up because medications are inherently compositional as a single drug product may contain multiple active ingredients, all of which may be clinically relevant. In contrast, SNOMED condition concepts are more naturally represented by a single higher-level ancestor. LOINC roll-up used the official LOINC group files, which provide predefined groupings of related laboratory tests and measurements sharing a common clinical purpose. In addition, numeric features were retained for LOINC events. These used raw LOINC codes rather than grouped or standardised concepts in order to avoid ambiguity introduced by grouping measurements with different units or scales. These were labelled as numeric columns. The benefit of this approach is that it preserves quantitative laboratory information where the absolute measured value may be clinically informative, while still allowing grouped representations for broader categorical modelling.

Events were then assigned to disjoint time bins corresponding to 0-30, 31-90, and 91-365 days before the scan cutoff. In addition, SNOMED and RxNorm features occurring more than 365 days before the scan cutoff were assigned to a separate "ever" bin to capture longer-term condition and medication history. This was not applied to LOINC features, as laboratory results are more time-sensitive and may lose clinical relevance over longer periods. Features were aggregated by group identifier and time window. For multiple

numeric LOINC features, the median value was used. After final merging, only binary features present in at least 2% of scans were retained. For numeric LOINC features, at least 3% of individuals within a given time window were required to have that feature, the feature had to be present in at least two time windows, and, where unit information was available, at least 95% of recorded values had to share the same dominant unit. These thresholds were applied to remove sparse, unstable, or semantically inconsistent events and to produce a more compact and robust EHR feature set. The final EHR feature matrix contained 1,597 features.

Two models, logistic regression and XGBoost [40], were used to evaluate the resulting EHR features. Logistic regression was included as a simple and interpretable linear baseline, providing a useful reference for whether the engineered EHR features were predictive under a relatively low-capacity model. XGBoost was included as a stronger non-linear model, as it can capture interactions and non-linear relationships between structured clinical features while remaining well suited to tabular data.

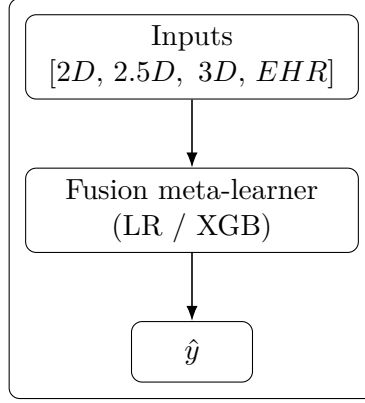
3.4 Ensemble and Fusion Strategy

Evaluation first considered the five base models individually in order to assess the standalone predictive value of each modality-specific approach. These comprised the three CT models and the two EHR models. CT ensemble modelling then examined three strategies for combining the predicted probabilities of the 2D, 2.5D, and 3D models. These were an unweighted mean ensemble, logistic regression as a meta-learner, and XGBoost as a meta-learner. The mean ensemble served as a simple non-learned baseline, while logistic regression and XGBoost provided learned ensemble methods of increasing complexity.

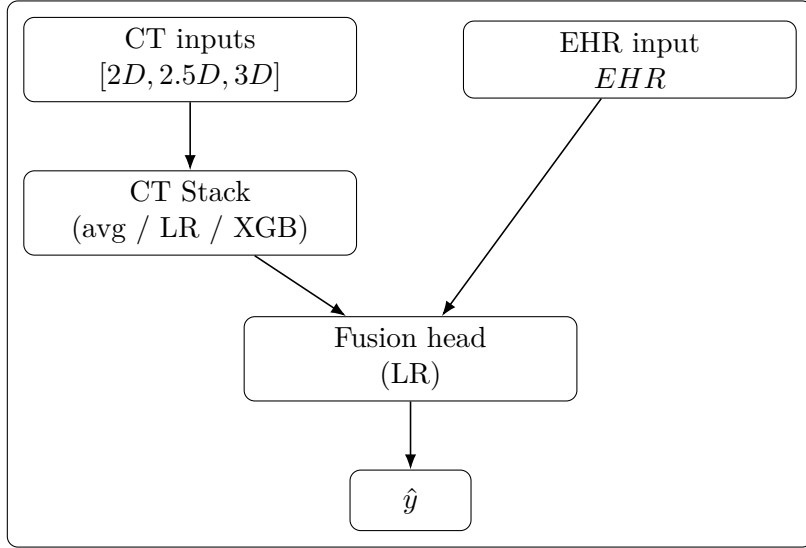
The effect of fusion timing was then evaluated using two multimodal strategies. In the flat fusion setting, the EHR model prediction was introduced at the same stage as CT stacking, so that all modality-specific predictions were combined jointly within a single meta-learning stage. In the hierarchical fusion setting, the CT model predictions were first combined into a CT ensemble, after which the resulting CT ensemble prediction was combined with the EHR model prediction in a second stage. This enabled the effect of introducing EHR information before or after modality-specific CT aggregation to be assessed directly.

For the flat fusion experiments, both logistic regression (LR) and eXtreme Gradient Boosting (XGB) were used as meta-learners with each of the two EHR models, yielding four multimodal model variants. For the hierarchical fusion experiments, each of the three CT ensemble models was paired with each EHR model, yielding six variants. At the final stage of hierarchical fusion, logistic regression was used as the combiner due to

the simplicity of only two modality-level predictions being merged at that stage.



(a) Flat late fusion.



(b) Hierarchical fusion.

Figure 3.3: Comparison of the two multimodal fusion strategies evaluated in this study. In flat fusion, the CT and EHR predictions are combined jointly within a single meta-learning stage. In hierarchical fusion, the CT predictions are first aggregated into a CT ensemble before being combined with the EHR prediction in a second stage.

3.5 Training

Training procedures differed between the CT and structured EHR models with model-specific optimisation settings summarised in Appendix A.2. For the CT models, early stopping was applied with a patience of 4 epochs based on validation AUPRC, and the checkpoint with the highest validation AUPRC was retained. AUPRC was used as the primary model-selection metric because both prediction tasks were class-imbalanced, making precision-recall performance more informative than accuracy for the positive class.

A `ReduceLROnPlateau` schedule was also applied, with reduction factor 0.5 and patience 1 epoch. Distributed training was used for the CT models across 3 GPUs. Gradient accumulation was used where necessary to accommodate memory constraints, with model-specific effective batch sizes given in Appendix A.2.

Cross-entropy loss was used as the training objective for the CT models. To account for class imbalance, class weights were set inversely proportional to the numbers of positive and negative samples in the training partition and normalised to sum to two. Optimisation used `AdamW` [41] with a maximum of 20 epochs, with model-specific learning rates, weight decay, and accumulation settings given in Appendix A.2. Thresholds were also computed on validation predictions for binary decision outputs as part of the implementation pipeline, but these outputs are not used in any of the subsequently reported results. The structured EHR models used separate training procedures appropriate for tabular learning, as detailed in Appendix A.2.

All stacked models were trained using strict 5-fold out-of-fold (OOF) stacking to prevent information leakage. Level-1 meta-learners were trained on fold-wise validation predictions from the base models, and hierarchical level-2 models were trained on out-of-fold predictions from the level-1 models, ensuring that no training input was generated by a model that had been fitted on the corresponding sample. The held-out test set was fixed across seeded runs, so variation in seed did not alter test-set membership. Instead, the seed affected stochastic components of model development, including parameter initialisation, optimisation randomness, and, for out-of-fold stages, the stratified fold assignment used to generate out-of-fold predictions. Seeds used are given in A.2. Test-set predictions were generated separately on this fixed held-out set and were never used during model fitting.

3.6 Evaluation Methods

Evaluation was conducted over five repeated runs, with all reported values presented as means across seeds with dispersion measure reported. Model performance was assessed using AUROC and AUPRC, with AUPRC treated as the primary discrimination metric because both prediction tasks were class-imbalanced. Calibration was evaluated descriptively using Brier score and expected calibration error (ECE), with no additional calibration procedure applied, as the aim was to compare model behaviour across architectures and fusion strategies rather than to optimise calibrated probabilities for deployment.

In addition to threshold-independent metrics, threshold-dependent behaviour was examined for the strongest flat and hierarchical fusion models on the test set. This was motivated by the clinical decision context, in which missed positive cases may be more

consequential than false positives and sensitivity may therefore be prioritised. Because AUROC and AUPRC do not show how models behave once a fixed decision threshold is applied, the threshold was varied directly on the test-set predictions to identify operating points achieving sensitivities of 85%, 90%, and 95%. This allowed the trade-off associated with increasingly sensitivity-favouring decision rules to be characterised.

Further analysis included branch-level SHAP comparisons for the strongest flat and hierarchical fusion models in order to assess the relative contribution of imaging and structured clinical information to the final prediction. Exact SHAP procedure is explained in A.3. Exploratory subgroup analysis was also performed for the strongest flat and hierarchical models across age bin, vendor group, and slice-thickness bin. These subgroup results are reported descriptively using AUROC and AUPRC, with strata compared to predefined stability thresholds, defined in A.3.

Chapter 4

Results

4.1 Single Modality Baselines

This section establishes the unimodal baselines for the study by evaluating CT-only models, EHR-only models, and CT ensemble models for both diagnosis and prognosis.

4.1.1 CT Single Models

This section evaluates the 2D, 2.5D, and 3D CT architectures for diagnosis and prognosis, comparing their baseline discrimination, calibration, and task-specific performance.

Table 4.1: Performance of individual CT-only models for diagnosis and prognosis. Values are reported as mean $\pm$ standard deviation across seeds.

Task	Model	AUROC	AUPRC	Brier	ECE
Diagnosis	2D	0.7603 $\pm$ 0.0041	0.6201 $\pm$ 0.0132	0.2041 $\pm$ 0.0142	0.1516 $\pm$ 0.0396
Diagnosis	2.5D	0.7741 $\pm$ 0.0102	0.6573 $\pm$ 0.0129	0.1836 $\pm$ 0.0063	0.1103 $\pm$ 0.0187
Diagnosis	3D	0.7434 $\pm$ 0.0144	0.5845 $\pm$ 0.0166	0.2108 $\pm$ 0.0318	0.1415 $\pm$ 0.0967
Prognosis	2D	0.7335 $\pm$ 0.0174	0.3028 $\pm$ 0.0179	0.1188 $\pm$ 0.0147	0.0884 $\pm$ 0.0479
Prognosis	2.5D	0.6711 $\pm$ 0.0643	0.2971 $\pm$ 0.0622	0.1467 $\pm$ 0.0212	0.1743 $\pm$ 0.0566
Prognosis	3D	0.6335 $\pm$ 0.0354	0.2083 $\pm$ 0.0329	0.2307 $\pm$ 0.0700	0.2732 $\pm$ 0.1345

The CT-only results indicate that performance depended on both task and image representation. Across all three architectures, performance was consistently stronger for diagnosis than for prognosis, particularly in AUPRC, indicating that prognosis was the more challenging prediction task. For diagnosis, 2.5D achieved the strongest overall performance, with an AUROC of 0.7741 and an AUPRC of 0.6573, and also showed the best calibration, with the lowest Brier score (0.1836) and ECE (0.1103). For prognosis, 2D produced the strongest overall performance, with an AUROC of 0.7335 and an AUPRC of 0.3028, while 3D performed worst on both tasks. Overall, these results indicate that

CT captures useful signal for both diagnosis and prognosis, but that the most effective image formulation differs between tasks.

4.1.2 EHR-only Models

This section evaluates the EHR-only models, logistic regression and extreme gradient boosting, for diagnosis and prognosis, comparing the relative performance of linear and non-linear approaches.

Table 4.2: Performance of individual EHR-only models for diagnosis and prognosis. Values are reported as means across seeds. Standard deviations are omitted because the EHR-only runs were deterministic across seeds.

Task	Model	AUROC	AUPRC	Brier	ECE
Diagnosis	LR	0.7383	0.5772	0.2009	0.1213
Diagnosis	XGB	0.8301	0.7063	0.1482	0.0616
Prognosis	LR	0.7198	0.3640	0.1349	0.1180
Prognosis	XGB	0.8630	0.5068	0.0886	0.0649

XGB consistently outperformed LR in the EHR-only setting for both diagnosis and prognosis. For diagnosis, AUROC increased from 0.7383 to 0.8301 and AUPRC from 0.5772 to 0.7063. The improvement was even larger for prognosis, where AUROC increased from 0.7198 to 0.8630 and AUPRC from 0.3640 to 0.5068. Calibration also improved under XGB in both tasks, with lower Brier score and ECE than LR. Notably, for prognosis, this EHR-only XGB baseline remained slightly stronger than any later multimodal configuration evaluated in this study, and therefore serves as the strongest overall prognosis reference point. These results indicate that structured clinical data alone provided substantial predictive signal for both tasks, and that this signal was better captured by a more flexible non-linear model than by a linear baseline, particularly for prognosis.

4.1.3 CT-only Ensemble Methods

This section evaluates CT-only ensemble models that combine the outputs of the individual CT architectures. The aim was to assess whether image-only ensembling improved performance relative to the single CT models, identify the strongest CT-only ensemble baseline for each task, and examine the contribution of the base CT model probabilities to the meta-learners using SHAP values.

The CT-only stacking results indicate that image-only ensembling provided, at most, modest gains over the individual CT baselines, and that these gains depended on both the task and the choice of meta-learner. For diagnosis, stacking slightly improved discrimination relative to the best single CT model, with CTMean achieving the highest

Table 4.3: Performance of CT-only stacking models for diagnosis and prognosis. Values are reported as mean $\pm$ standard deviation across seeds.

Task	Model	AUROC	AUPRC	Brier	ECE
Diagnosis	CTLR	0.7838 ± 0.0098	0.6681 ± 0.0097	0.1686 ± 0.0024	0.0484 ± 0.0174
Diagnosis	CTMean	0.7865 ± 0.0093	0.6642 ± 0.0105	0.1844 ± 0.0066	0.1298 ± 0.0293
Diagnosis	CTXGB	0.7713 ± 0.0124	0.6433 ± 0.0170	0.1718 ± 0.0036	0.0490 ± 0.0041
Prognosis	CTLR	0.7280 ± 0.0244	0.2871 ± 0.0332	0.1062 ± 0.0028	0.0379 ± 0.0127
Prognosis	CTMean	0.7106 ± 0.0310	0.2709 ± 0.0352	0.1420 ± 0.0191	0.1638 ± 0.0591
Prognosis	CTXGB	0.6873 ± 0.0517	0.2712 ± 0.0591	0.1116 ± 0.0068	0.0597 ± 0.0174

AUROC (0.7865 vs 0.7741 for the best single 2.5D model), while CTLR achieved the highest AUPRC (0.6681) and the strongest calibration (Brier 0.1686, ECE 0.0484). This suggests that ensembling was beneficial for diagnosis, although no single stacker was uniformly best across all metrics. For prognosis, CTLR was the strongest image-only ensemble, with an AUROC of 0.7280 and an AUPRC of 0.2871, but this did not exceed the best single CT baseline for discrimination, indicating that stacking offered limited benefit for ranking performance on this task. Its main advantage in prognosis was improved calibration, with lower Brier score and ECE than the single-model baselines.

Table 4.4: Mean absolute SHAP values for the learned CT-stacking models. Values are reported as mean $\pm$ standard deviation across seeds.

Task	CT stacker	2D	2.5D	3D
Diagnosis	CTLR	0.3838 ± 0.0538	0.3741 ± 0.0392	0.1165 ± 0.0638
Diagnosis	CTXGB	0.4493 ± 0.0875	0.4161 ± 0.0476	0.1575 ± 0.0605
Prognosis	CTLR	0.2589 ± 0.0170	0.2437 ± 0.1598	0.1422 ± 0.0490
Prognosis	CTXGB	0.5297 ± 0.1223	0.5540 ± 0.3057	0.3756 ± 0.0953

SHAP analysis showed that the CT stackers relied primarily on the 2D and 2.5D base-model probabilities, while the 3D branch contributed less overall. In diagnosis, for example, CTLR assigned similar importance to 2D and 2.5D (0.3838 and 0.3741), compared with 0.1165 for 3D. A similar pattern was observed in prognosis, although CTXGB attributed somewhat more importance to the 3D branch. Overall, these results suggest that the value of CT stacking arose mainly from combining the stronger 2D and 2.5D representations.

4.1.4 Synthesis

Overall, the single-modality baselines showed that EHR-only models achieved stronger standalone performance than CT-only models for both diagnosis and prognosis. By contrast, CT-only stacking produced only limited changes relative to the best individual

CT models, with modest improvement for diagnosis and less clear benefit for prognosis. These results establish the single-modality reference points against which the multimodal models can be compared.

4.2 Comparison of Flat and Hierarchical Fusion for Diagnosis

4.2.1 Flat fusion models for diagnosis

This subsection evaluates the flat multimodal fusion models for diagnosis, comparing performance across EHR branches and meta-learners. In the flat setting, CT-derived and EHR-derived information are combined at the same stage.

Table 4.5: Performance of flat multimodal fusion models for diagnosis. Values are reported as mean $\pm$ standard deviation across seeds.

EHR model	Meta model	AUROC	AUPRC	Brier	ECE
LR	LR	0.7938 ± 0.0066	0.6607 ± 0.0053	0.1591 ± 0.0018	0.0386 ± 0.0090
LR	XGB	0.7942 ± 0.0120	0.6557 ± 0.0128	0.1606 ± 0.0026	0.0363 ± 0.0063
XGB	LR	0.8470 ± 0.0033	0.7288 ± 0.0024	0.1358 ± 0.0007	0.0618 ± 0.0048
XGB	XGB	0.8405 ± 0.0077	0.7157 ± 0.0178	0.1371 ± 0.0031	0.0449 ± 0.0109

In terms of discrimination, the flat diagnosis models showed a clear split by EHR branch. Models using the XGB EHR branch outperformed those using the LR EHR branch, with AUROC increasing from 0.7938-0.7942 to 0.8405-0.8470 and AUPRC from 0.6557-0.6607 to 0.7157-0.7288. The strongest discriminative performance was achieved by the XGB-EHR + LR-meta model, which reached an AUROC of 0.8470 and an AUPRC of 0.7288. This indicates that, for diagnosis, the choice of EHR model had a larger effect on discrimination than the choice of meta-learner.

The calibration pattern was less closely aligned with discrimination. The two models using the LR EHR branch achieved the lowest ECE values, with LR-EHR + XGB-meta performing best overall on this metric at 0.0363. By contrast, the strongest discriminative model, XGB-EHR + LR-meta, achieved the lowest Brier score (0.1358) but a higher ECE (0.0618). Taken together, these results indicate that, within the flat diagnosis models, discrimination varied more with the EHR branch than with the meta-learner, while calibration differences were smaller and less consistent.

4.2.2 Hierarchical fusion models for diagnosis

This subsection evaluates the hierarchical multimodal fusion models for diagnosis, comparing performance across CT stackers and EHR branches. In the hierarchical setting,

CT-derived information is first combined through a CT-level ensemble before being integrated with the EHR branch.

Table 4.6: Performance of hierarchical multimodal fusion models for diagnosis. Values are reported as mean $\pm$ standard deviation across seeds.

CT model	EHR model	Meta model	AUROC	AUPRC	Brier	ECE
CTLR	LR	LR	0.7895 $\pm$ 0.0063	0.6566 $\pm$ 0.0038	0.1601 $\pm$ 0.0017	0.0383 $\pm$ 0.0111
CTLR	XGB	LR	0.8453 $\pm$ 0.0031	0.7273 $\pm$ 0.0021	0.1366 $\pm$ 0.0005	0.0657 $\pm$ 0.0064
CTMean	LR	LR	0.7936 $\pm$ 0.0074	0.6553 $\pm$ 0.0059	0.1599 $\pm$ 0.0018	0.0369 $\pm$ 0.0053
CTMean	XGB	LR	0.8474 $\pm$ 0.0034	0.7266 $\pm$ 0.0039	0.1365 $\pm$ 0.0006	0.0618 $\pm$ 0.0047
CTXGB	LR	LR	0.7849 $\pm$ 0.0072	0.6516 $\pm$ 0.0132	0.1625 $\pm$ 0.0025	0.0390 $\pm$ 0.0057
CTXGB	XGB	LR	0.8446 $\pm$ 0.0033	0.7264 $\pm$ 0.0068	0.1377 $\pm$ 0.0009	0.0598 $\pm$ 0.0066

In terms of discrimination, the hierarchical diagnosis models showed the same broad pattern as the flat models: configurations using the XGB EHR branch outperformed those using the LR EHR branch. AUROC increased from 0.7849–0.7936 for the LR-EHR models to 0.8446–0.8474 for the XGB-EHR models, while AUPRC increased from 0.6516–0.6566 to 0.7264–0.7273. Within the hierarchical family, the strongest overall discriminative performance was achieved by the CTMean + XGB-EHR + LR-meta configuration, which reached the highest AUROC of 0.8474, while CTLR + XGB-EHR + LR-meta achieved the highest AUPRC of 0.7273. Overall, these results indicate that the choice of EHR branch had a larger effect on discrimination than the choice of CT stacker.

The calibration pattern was again less closely aligned with discrimination. Among the LR-EHR hierarchical models, CTMean + LR-EHR + LR-meta achieved the lowest ECE at 0.0369, whereas the strongest discriminative configurations had higher ECE values, ranging from 0.0598 to 0.0657. At the same time, CTMean + XGB-EHR + LR-meta achieved the lowest Brier score overall at 0.1365, indicating that the strongest hierarchical model by discrimination was also strongest on this calibration metric, though not on ECE. These results indicate that, within the hierarchical diagnosis models, discrimination varied more with the EHR branch than with the CT stacker, while calibration again did not identify a single uniformly dominant configuration.

4.2.3 Comparison of strongest flat and hierarchical models for diagnosis

Having evaluated the flat and hierarchical diagnosis families separately, one representative model from each was selected for direct comparison. Restricting the comparison to these models reduced variation arising from within-family design choices and allowed the analysis to focus on whether the stage of EHR integration affected multimodal diagnosis performance. The comparison considered discrimination, calibration, and SHAP-based

feature importance to assess both predictive performance and the relative contribution of the CT and EHR branches.

Within the flat family, the XGB-EHR + LR-meta model was selected as the highest performing configuration with superior discrimination values and Brier score. Within the hierarchical family, the CTMean + XGB-EHR + LR-meta model was carried forward as the strongest overall configuration, achieving the highest AUROC and lowest Brier score, while remaining effectively tied on AUPRC with the top hierarchical alternative.

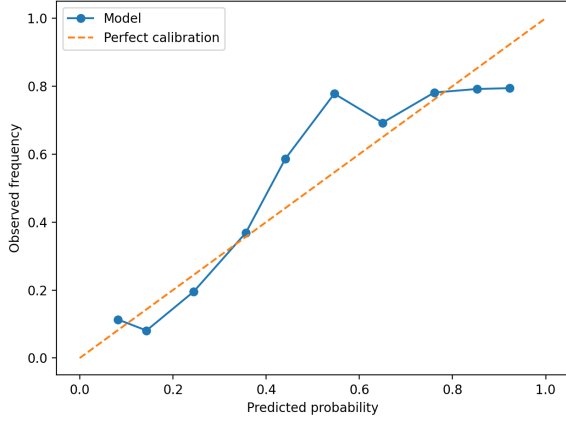
Table 4.7: Comparison of the selected flat and hierarchical multimodal fusion models for diagnosis. Values are reported as mean $\pm$ standard deviation across seeds.

Metric	Flat: XGB-EHR + LR-meta	Hierarchical: CTMean + XGB-EHR + LR-meta
AUROC	0.8470 $\pm$ 0.0033	0.8474 $\pm$ 0.0034
AUPRC	0.7288 $\pm$ 0.0024	0.7266 $\pm$ 0.0039
Brier	0.1358 $\pm$ 0.0007	0.1365 $\pm$ 0.0006
ECE	0.0618 $\pm$ 0.0048	0.0618 $\pm$ 0.0047

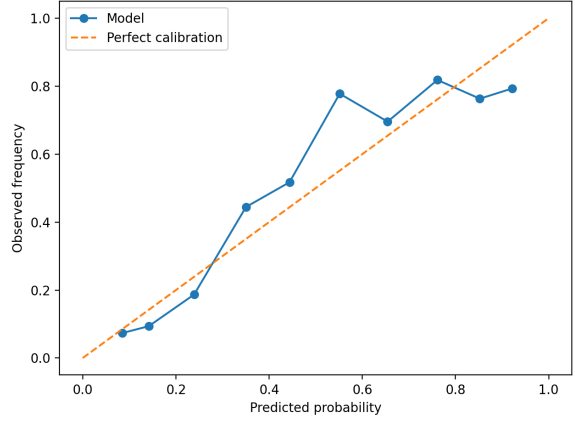
The strongest flat and hierarchical diagnosis models showed near-identical performance. The best flat model achieved an AUROC of 0.8470 and an AUPRC of 0.7288, while the best hierarchical model achieved an AUROC of 0.8474 and an AUPRC of 0.7266. This indicates that both fusion strategies extracted a similarly strong multimodal diagnostic signal, with only negligible differences in overall discrimination. The hierarchical model had a marginally higher AUROC, whereas the flat model had a marginally higher AUPRC, but these differences were too small to support any strong claim of discriminative superiority for either approach.

Calibration was similarly comparable between the two models. The flat model achieved a slightly lower Brier score of 0.1358, compared with 0.1365 for the hierarchical model, while both models had the same ECE of 0.0618. Combined, these findings suggest that, for diagnosis, multimodal performance was more influenced by combining CT and EHR than by the precise stage at which EHR was introduced. In practical terms, flat and hierarchical fusion performed equivalently for this task, both in discrimination and calibration.

The SHAP profiles of the strongest diagnosis models showed that both fusion strategies were driven primarily by the EHR branch, with CT making a smaller secondary contribution. In the strongest flat model, the EHR XGB branch had the largest mean absolute SHAP value (0.9431), while the CT contribution was distributed across the 2.5D, 2D, and 3D branches, with 2.5D contributing most. A similar pattern was observed in the strongest hierarchical model, where the EHR XGB branch (0.9568) contributed more than the aggregated CTMean branch (0.3924). Overall, these results indicate that at



(a) Selected flat diagnosis model (XGB-EHR + LR-meta).



(b) Selected hierarchical diagnosis model (CT-Mean + XGB-EHR + LR-meta).

Figure 4.1: Calibration curves for the selected flat and hierarchical diagnosis fusion models, shown for seed 1, which was the most representative seed for both models based on overall performance relative to the across-seed mean.

Table 4.8: Mean absolute SHAP values for the selected diagnosis fusion models. Values are reported as mean $\pm$ standard deviation across seeds. Higher values indicate greater contribution of the corresponding branch to the final stacked prediction.

Fusion strategy	Branch	Mean absolute SHAP
Flat	2D	0.1812 ± 0.0567
Flat	2.5D	0.2260 ± 0.0390
Flat	3D	0.0689 ± 0.0364
Flat	EHR (XGB)	0.9431 ± 0.0633
Hierarchical	CT stacker (CTMean)	0.3924 ± 0.0279
Hierarchical	EHR (XGB)	0.9568 ± 0.0621

the final fusion stage, both selected diagnosis models placed greater importance on the EHR branch output than on the CT-derived branch outputs. Differences between flat and hierarchical fusion arising mainly from how CT information was organised rather than from the relative importance of EHR.

Because both selected diagnosis models used the XGB EHR branch, a secondary matched comparison was performed using the highest-AUPRC LR-EHR configurations in each family. This was LR-EHR + LR-meta in the flat setting and CTRLR + LR-EHR + LR-meta in the hierarchical setting. These models again performed very similarly, with AUROC values of 0.7938 and 0.7895, AUPRC values of 0.6607 and 0.6566, Brier scores of 0.1591 and 0.1601, and ECE values of 0.0386 and 0.0383, respectively. SHAP summaries for this matched comparison also assigned greater contribution to the EHR branch than to the CT inputs. In the flat LR-EHR model, the EHR branch had the largest mean absolute SHAP value (0.6837), with smaller contributions from 2D (0.2549), 2.5D (0.2291), and 3D (0.0852). In the hierarchical LR-EHR model, the EHR branch (0.6978) contributed more than the aggregated CTRLR branch (0.4549) but with a closer balance than seen with XGB models. Overall, these results indicate that the limited difference between flat and hierarchical fusion for diagnosis was also evident in a matched LR-EHR setting, where the EHR contribution was more closely balanced with CT than in the stronger XGB-EHR comparison. This reduces the concern that the near-equivalence between fusion strategies was specific to the strongly weighted XGB-EHR model.

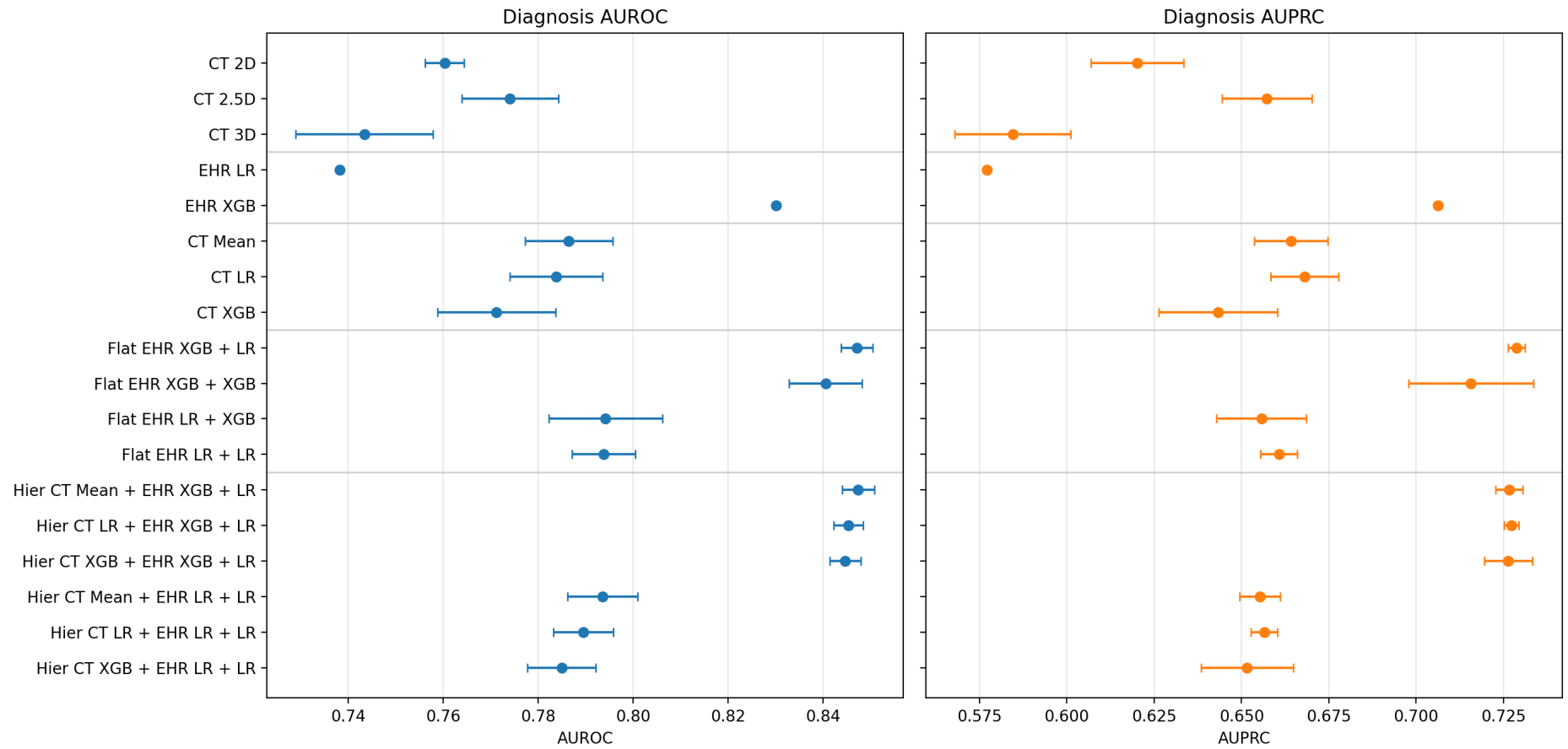


Figure 4.2: Overview of AUROC and AUPRC across all evaluated diagnostic models. Points show mean performance across seeds, with error bars indicating standard deviation.

4.3 Comparison of Flat and Hierarchical Fusion for Prognosis

4.3.1 Flat fusion models for prognosis

This subsection evaluates the flat multimodal fusion models for prognosis, comparing performance across EHR branches and meta-learners.

Table 4.9: Performance of flat multimodal fusion models for prognosis. Values are reported as mean $\pm$ standard deviation across seeds.

EHR model	Meta model	AUROC	AUPRC	Brier	ECE
LR	LR	0.7823 $\pm$ 0.0132	0.3912 $\pm$ 0.0250	0.0991 $\pm$ 0.0019	0.0465 $\pm$ 0.0116
LR	XGB	0.7497 $\pm$ 0.0429	0.3176 $\pm$ 0.0532	0.1075 $\pm$ 0.0046	0.0546 $\pm$ 0.0136
XGB	LR	0.8349 $\pm$ 0.0120	0.4707 $\pm$ 0.0124	0.0908 $\pm$ 0.0012	0.0384 $\pm$ 0.0156
XGB	XGB	0.8054 $\pm$ 0.0306	0.3824 $\pm$ 0.0361	0.1028 $\pm$ 0.0082	0.0579 $\pm$ 0.0185

In terms of discrimination, the flat prognosis models again showed a clear split by EHR branch. Models using the XGB EHR branch outperformed those using the LR EHR branch, with AUROC increasing from 0.7497–0.7823 to 0.8054–0.8349 and AUPRC from 0.3176–0.3912 to 0.3824–0.4707. The strongest discriminative performance was achieved by the XGB-EHR + LR-meta model, which reached an AUROC of 0.8349 and an AUPRC of 0.4707. Unlike in diagnosis, the choice of meta-learner also had a clearer effect in prognosis, with the LR meta-learner outperforming the XGB meta-learner across both EHR branches.

The calibration pattern was more closely aligned with discrimination than in diagnosis. The XGB-EHR + LR-meta model achieved the lowest Brier score (0.0908) and the lowest ECE (0.0384), making it the strongest flat prognosis model on both discrimination and calibration. By contrast, the XGB-meta configurations were weaker across both types of metric, with higher Brier scores and ECE values under both the LR and XGB EHR branches. These results indicate that, within the flat prognosis models, the XGB-EHR + LR-meta configuration was clearly dominant, and that meta-learner choice had a more visible effect on performance than in the diagnosis setting.

4.3.2 Hierarchical fusion models for prognosis

This subsection evaluates the hierarchical multimodal fusion models for prognosis, comparing performance across CT stackers and EHR branches.

In terms of discrimination, the hierarchical prognosis models again showed a clear split by EHR branch. Configurations using the XGB EHR branch outperformed those us-

Table 4.10: Performance of hierarchical multimodal fusion models for prognosis. Values are reported as mean $\pm$ standard deviation across seeds.

CT model	EHR model	Meta model	AUROC	AUPRC	Brier	ECE
CTLR	LR	LR	0.7595 $\pm$ 0.0171	0.3980 $\pm$ 0.0146	0.1004 $\pm$ 0.0016	0.0396 $\pm$ 0.0104
CTLR	XGB	LR	0.8622 $\pm$ 0.0303	0.4995 $\pm$ 0.0521	0.0908 $\pm$ 0.0081	0.0769 $\pm$ 0.0300
CTMean	LR	LR	0.7779 $\pm$ 0.0130	0.3843 $\pm$ 0.0186	0.0999 $\pm$ 0.0016	0.0373 $\pm$ 0.0114
CTMean	XGB	LR	0.8302 $\pm$ 0.0207	0.4724 $\pm$ 0.0525	0.0915 $\pm$ 0.0080	0.0480 $\pm$ 0.0164
CTXGB	LR	LR	0.7606 $\pm$ 0.0140	0.3793 $\pm$ 0.0335	0.1008 $\pm$ 0.0029	0.0346 $\pm$ 0.0086
CTXGB	XGB	LR	0.8514 $\pm$ 0.0228	0.4920 $\pm$ 0.0532	0.0910 $\pm$ 0.0074	0.0698 $\pm$ 0.0218

ing the LR EHR branch, with AUROC increasing from 0.7595–0.7779 to 0.8302–0.8622 and AUPRC from 0.3793–0.3980 to 0.4724–0.4995. Within the hierarchical family, the strongest discriminative performance was achieved by the CTLR + XGB-EHR + LR-meta configuration, which reached an AUROC of 0.8622 and an AUPRC of 0.4995. Unlike the diagnosis setting, variation across CT stackers was also more pronounced here, with CTLR performing best, CTXGB close behind, and CTMean weaker when paired with the XGB EHR branch.

The calibration pattern was less closely aligned with discrimination. The strongest discriminative model, CTLR + XGB-EHR + LR-meta, also achieved the lowest Brier score (0.0908), but it did not achieve the lowest ECE. Instead, the lowest ECE was observed for CTXGB + LR-EHR + LR-meta at 0.0346, followed closely by CTMean + LR-EHR + LR-meta at 0.0373. This indicates that, within the hierarchical prognosis models, the configurations with the best ranking performance were not also the best calibrated under every calibration metric. Overall, these results indicate that, within the hierarchical prognosis models, discrimination still differed more by EHR branch than by CT stacker, although CT stacker choice had a more visible effect than in diagnosis, while calibration again did not identify a single uniformly dominant configuration.

4.3.3 Comparison of strongest flat and hierarchical models for prognosis

Having evaluated the flat and hierarchical prognosis families separately, one representative model from each was selected for direct comparison. Restricting the comparison to these models reduced variation from within-family design choices and allowed the analysis to focus on whether the stage of EHR integration affected prognostic performance. The comparison considered discrimination, calibration, and SHAP-based feature importance.

The selected flat model was XGB-EHR + LR-meta, and the selected hierarchical model was CTLR + XGB-EHR + LR-meta.

The selected hierarchical and flat prognosis models were more clearly separated than the

Table 4.11: Comparison of the selected flat and hierarchical multimodal fusion models for prognosis. Values are reported as mean $\pm$ standard deviation across seeds.

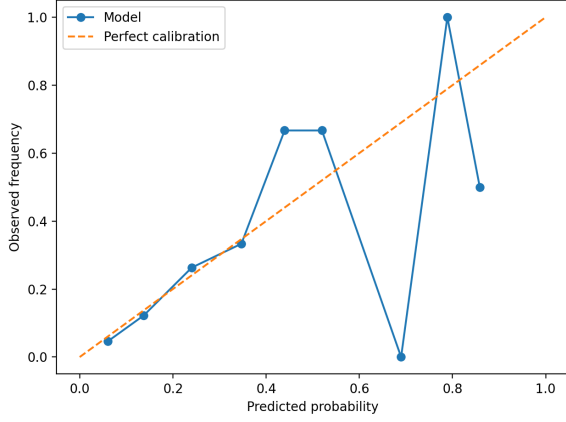
Metric	Flat: XGB-EHR + LR-meta	Hierarchical: CTRLR + XGB-EHR + LR-meta
AUROC	0.8349 $\pm$ 0.0120	0.8622 $\pm$ 0.0303
AUPRC	0.4707 $\pm$ 0.0124	0.4995 $\pm$ 0.0521
Brier	0.0908 $\pm$ 0.0012	0.0908 $\pm$ 0.0081
ECE	0.0384 $\pm$ 0.0156	0.0769 $\pm$ 0.0300

corresponding diagnosis models. The best flat model achieved an AUROC of 0.8349 and an AUPRC of 0.4707, whereas the best hierarchical model achieved an AUROC of 0.8622 and an AUPRC of 0.4995. This indicates that, in the selected configurations, hierarchical fusion provided stronger prognostic discrimination than the flat alternative. Unlike the diagnosis comparison, where the two fusion strategies were effectively equivalent, the prognosis results suggest a more meaningful advantage for hierarchical fusion in ranking performance. However, neither selected multimodal prognosis model exceeded the EHR-only XGB baseline reported in Table 4.2. Accordingly, this comparison should be interpreted as evidence that the stage of EHR integration affected performance within CT-augmented multimodal models, rather than as evidence that CT added net prognostic value beyond the strongest structured-data model.

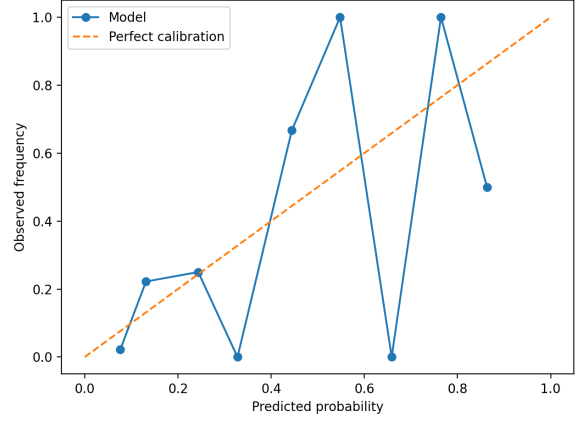
The calibration comparison was more mixed. Both models achieved the same Brier score of 0.0908, indicating comparable overall probability error, but the flat model had a substantially lower ECE (0.0384) than the hierarchical model (0.0769). This suggests that although the hierarchical model discriminated better, the flat model was better calibrated by ECE. Taken together, these findings indicate that, for prognosis, the stage of EHR integration appeared to matter more than it did for diagnosis. Hierarchical fusion improved discrimination, but this did not translate into uniformly better calibration across all metrics.

The calibration curves in Figure 4.3 are consistent with the metric-level comparison. The flat model follows the diagonal more closely overall, in line with its lower ECE, whereas the hierarchical model shows larger local deviations despite matching on Brier score. Given the small prognosis test set, these curves should be interpreted qualitatively.

The SHAP profiles of the selected prognosis models showed that both fusion strategies relied more heavily on the EHR branch than on the CT-derived branch. In the selected flat model, the EHR XGB branch had the largest mean absolute SHAP value (0.4311), while the CT contribution was distributed across the 2D (0.2241), 2.5D (0.2131), and 3D (0.0849) branches. This indicates that the flat model still made use of multiple CT representations alongside EHR. In the selected hierarchical model, the EHR XGB branch (0.4938) contributed more than the aggregated CTRLR branch (0.1591), suggesting a more



(a) Selected flat prognosis model (XGB-EHR + LR-meta), shown for seed 42.



(b) Selected hierarchical prognosis model (CTLR + XGB-EHR + LR-meta), shown for seed 0.

Figure 4.3: Calibration curves for the selected flat and hierarchical prognosis fusion models. Both panels show illustrative single-seed calibration behaviour; mean calibration performance across seeds is reported in Table 4.11.

Table 4.12: Mean absolute SHAP values for the selected prognosis fusion models. Values are reported as mean $\pm$ standard deviation across seeds. Higher values indicate greater contribution of the corresponding branch to the final stacked prediction.

Fusion strategy	Branch	Mean absolute SHAP
Flat	2D	0.2241 ± 0.0313
Flat	2.5D	0.2131 ± 0.1453
Flat	3D	0.0849 ± 0.0470
Flat	EHR (XGB)	0.4311 ± 0.0434
Hierarchical	CT stacker (CTLR)	0.1591 ± 0.0446
Hierarchical	EHR (XGB)	0.4938 ± 0.0183

concentrated reliance on EHR at the final fusion stage. Taken together, these results indicate that prognosis remained primarily EHR-led in both selected fusion strategies, and that the stronger discrimination of the hierarchical model did not coincide with greater final-stage reliance on CT, and is therefore more consistent with differences in how CT information was organised before final fusion.

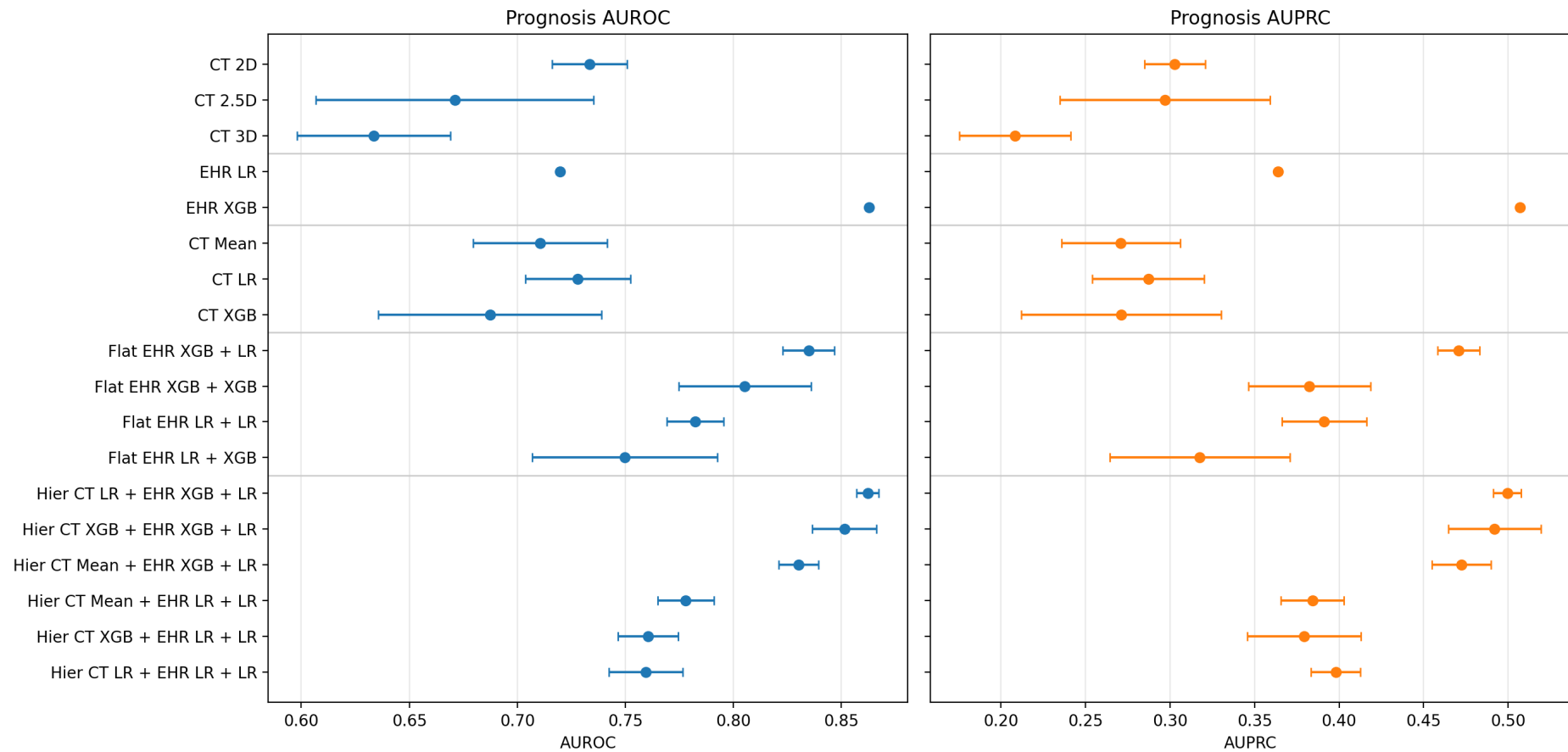


Figure 4.4: Overview of AUROC and AUPRC across all evaluated prognostic models. Points show mean performance across seeds, with error bars indicating standard deviation.

4.4 Operating Point Analysis

While AUROC, AUPRC, and calibration summarise overall model performance, clinical use ultimately requires a fixed decision threshold. This section therefore examines operating-point behaviour for the selected flat and hierarchical fusion models carried forward from the preceding comparisons, assessing how they trade sensitivity against specificity under increasingly sensitivity-favouring decision rules.

4.4.1 Diagnosis

For diagnosis, the same selected flat and hierarchical models carried forward in Section 4.2.3 were compared at test-set operating points corresponding to target sensitivities of 85%, 90%, and 95%.

Table 4.13: Test-set operating-point analysis for the selected flat diagnosis fusion model, XGB-EHR + LR-meta, at target sensitivities of 85%, 90%, and 95%. Values are reported as mean $\pm$ standard deviation across seeds.

Target sensitivity	Threshold	Achieved sensitivity	Specificity	PPV
85%	0.200 $\pm$ 0.016	0.856 $\pm$ 0.004	0.693 $\pm$ 0.031	0.569 $\pm$ 0.024
90%	0.150 $\pm$ 0.017	0.905 $\pm$ 0.003	0.515 $\pm$ 0.055	0.470 $\pm$ 0.029
95%	0.094 $\pm$ 0.009	0.964 $\pm$ 0.011	0.187 $\pm$ 0.039	0.359 $\pm$ 0.009

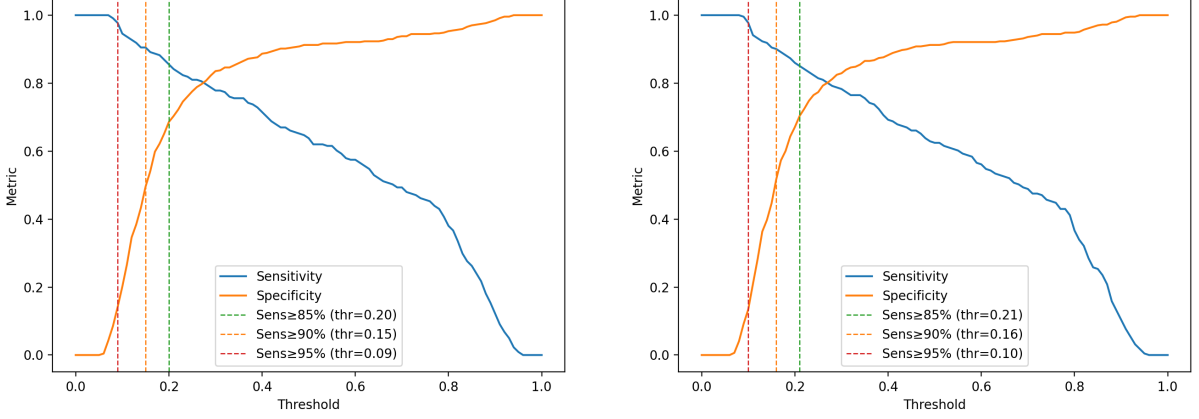
Table 4.14: Test-set operating-point analysis for the selected hierarchical diagnosis fusion model, CTMean + XGB-EHR + LR-meta, at target sensitivities of 85%, 90%, and 95%. Values are reported as mean $\pm$ standard deviation across seeds.

Target sensitivity	Threshold	Achieved sensitivity	Specificity	PPV
85%	0.206 $\pm$ 0.017	0.854 $\pm$ 0.006	0.714 $\pm$ 0.012	0.585 $\pm$ 0.011
90%	0.156 $\pm$ 0.022	0.901 $\pm$ 0.002	0.541 $\pm$ 0.056	0.483 $\pm$ 0.032
95%	0.100 $\pm$ 0.012	0.958 $\pm$ 0.011	0.203 $\pm$ 0.052	0.363 $\pm$ 0.013

For both selected diagnosis models, progressively higher sensitivity targets produced the expected operating-point trade-off. As the target increased from 85% to 95%, the decision threshold decreased, achieved sensitivity rose, and both specificity and Positive Predictive Value (PPV) fell. In the flat model, specificity declined from 0.693 to 0.187 and PPV from 0.569 to 0.359, while in the hierarchical model specificity declined from 0.714 to 0.203 and PPV from 0.585 to 0.363. This indicates that increasingly sensitivity-favouring decision rules substantially reduced rule-out precision and specificity in both configurations.

The comparison between models remained consistent across the three operating points. At matched sensitivity targets, the hierarchical model used slightly higher thresholds and achieved marginally higher specificity and PPV than the flat model, while maintaining

very similar achieved sensitivity. For example, at the 90% target, the hierarchical model achieved sensitivity 0.901 with specificity 0.541 and PPV 0.483, compared with 0.905, 0.515, and 0.470 for the flat model. Similar small advantages for the hierarchical model were present at the 85% and 95% targets. Overall, these results suggest that although the two diagnosis fusion strategies were very similar in overall discrimination, the hierarchical model provided a slightly more favourable sensitivity–specificity trade-off across the operating points examined.



(a) Selected flat diagnosis model (XGB-EHR + LR-meta), shown for seed 1.

(b) Selected hierarchical diagnosis model (CT-Mean + XGB-EHR + LR-meta), shown for seed 1.

Figure 4.5: Sensitivity and specificity as functions of the decision threshold for the selected diagnosis fusion models. Dashed lines mark the thresholds corresponding to target sensitivities of 85%, 90%, and 95%. The two models show very similar operating characteristics, with the hierarchical model achieving slightly higher thresholds and marginally better specificity and PPV at matched sensitivity targets

4.4.2 Prognosis

For prognosis, the same selected flat and hierarchical models carried forward in Section 4.3.3 were compared at test-set operating points corresponding to target sensitivities of 85%, 90%, and 95%.

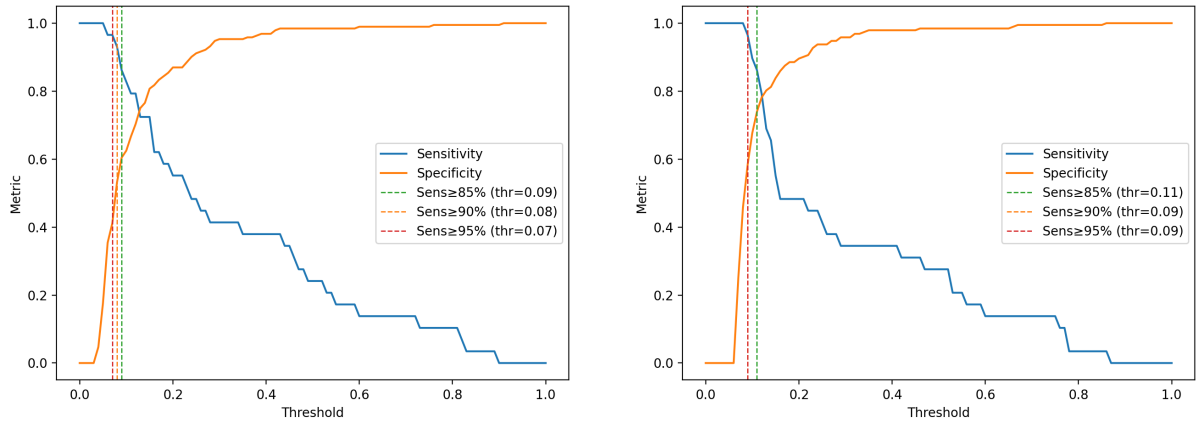
Table 4.15: Test-set operating-point analysis for the selected flat prognosis fusion model, XGB-EHR + LR-meta, at target sensitivities of 85%, 90%, and 95%. Values are reported as mean $\pm$ standard deviation across seeds.

Target sensitivity	Threshold	Achieved sensitivity	Specificity	PPV
85%	0.086 $\pm$ 0.011	0.869 $\pm$ 0.015	0.598 $\pm$ 0.056	0.248 $\pm$ 0.027
90%	0.074 $\pm$ 0.013	0.945 $\pm$ 0.019	0.494 $\pm$ 0.090	0.223 $\pm$ 0.032
95%	0.066 $\pm$ 0.018	0.979 $\pm$ 0.019	0.384 $\pm$ 0.149	0.200 $\pm$ 0.043

Table 4.16: Test-set operating-point analysis for the selected hierarchical prognosis fusion model, CTRLR + XGB-EHR + LR-meta, at target sensitivities of 85%, 90%, and 95%. Values are reported as mean $\pm$ standard deviation across seeds.

Target sensitivity	Threshold	Achieved sensitivity	Specificity	PPV
85%	0.104 ± 0.009	0.883 ± 0.031	0.711 ± 0.041	0.318 ± 0.025
90%	0.092 ± 0.008	0.952 ± 0.019	0.621 ± 0.064	0.278 ± 0.031
95%	0.088 ± 0.004	0.966 ± 0.000	0.578 ± 0.007	0.257 ± 0.003

Across both selected prognosis models, increasing the sensitivity target from 85% to 95% lowered the decision threshold and reduced specificity and PPV, as expected under increasingly sensitivity-favouring decision rules. However, the hierarchical model retained a more favourable trade-off at every operating point. At the 90% target, for example, it achieved sensitivity 0.952 with specificity 0.621 and PPV 0.278, compared with 0.945, 0.494, and 0.223 for the flat model. The same pattern was present at the 85% and 95% targets, where the hierarchical model consistently maintained higher specificity and PPV. These results indicate that the hierarchical prognosis model remained better balanced under high-sensitivity thresholds, reinforcing the advantage already seen in its overall discrimination.



(a) Selected flat prognosis model (XGB-EHR + LR-meta), shown for seed 0.

(b) Selected hierarchical prognosis model (CTRLR + XGB-EHR + LR-meta), shown for seed 0.

Figure 4.6: Sensitivity and specificity as functions of the decision threshold for the selected prognosis fusion models. Dashed lines mark the thresholds corresponding to target sensitivities of 85%, 90%, and 95%.

4.5 Subgroup Comparative Analysis

To assess whether the relative performance of flat and hierarchical fusion varied across clinically relevant strata, an exploratory subgroup analysis was performed for diagno-

sis only. Prognosis was not included because none of the prognosis subgroup bins met the predefined stability thresholds, making descriptive comparison unreliable. Subgroup analysis was restricted to the two best diagnosis models selected for the main comparison.

Table 4.17: Subgroup analysis for diagnosis by age bin. Values are reported as mean $\pm$ standard deviation across seeds. Positive deltas indicate better performance for hierarchical fusion.

Age bin	N	AUROC			AUPRC		
		Flat	Hierarchical	Δ	Flat	Hierarchical.	Δ
40–54	138	0.8241 $\pm$ 0.0057	0.8260 $\pm$ 0.0074	0.0019	0.6346 $\pm$ 0.0077	0.6367 $\pm$ 0.0067	0.0021
55–69	233	0.8350 $\pm$ 0.0055	0.8354 $\pm$ 0.0043	0.0005	0.7765 $\pm$ 0.0030	0.7734 $\pm$ 0.0035	-0.0031
70–84	186	0.8749 $\pm$ 0.0043	0.8746 $\pm$ 0.0050	-0.0003	0.7741 $\pm$ 0.0032	0.7742 $\pm$ 0.0079	0.0000

After excluding the unstable 0–39 age bin, the remaining age strata showed only very small differences between the flat and hierarchical diagnosis models. These deltas were negligible in magnitude and generally small relative to the across-seed variation, with only a marginal hierarchical advantage in the 40–54 group and near-identical performance in the older age bins. Overall, the age subgroup analysis does not indicate a meaningful age-dependent difference in the relative performance of the two fusion strategies.

Table 4.18: Subgroup analysis for diagnosis by scanner vendor group. Values are reported as mean $\pm$ standard deviation across seeds. Positive deltas indicate better performance for hierarchical fusion.

Vendor group	N	AUROC			AUPRC		
		Flat	Hier.	Δ	Flat	Hier.	Δ
GE	248	0.7806 $\pm$ 0.0086	0.7834 $\pm$ 0.0074	0.0028	0.6168 $\pm$ 0.0095	0.6147 $\pm$ 0.0078	-0.0021
TOSHIBA/CANON	204	0.6498 $\pm$ 0.0130	0.6494 $\pm$ 0.0118	-0.0004	0.2628 $\pm$ 0.0165	0.2601 $\pm$ 0.0141	-0.0026
Unknown	184	0.6051 $\pm$ 0.0059	0.5975 $\pm$ 0.0074	-0.0077	0.8042 $\pm$ 0.0032	0.8007 $\pm$ 0.0043	-0.0036

After excluding the unstable SIEMENS vendor group, differences between the flat and hierarchical diagnosis models were small within each retained vendor stratum. The more meaningful signal was instead between vendor groups, as absolute performance varied much more across vendors than the flat–hierarchical deltas did within vendors. This suggests that vendor group influenced overall subgroup difficulty more than it influenced the relative benefit of either fusion strategy

Across the revised slice-thickness bins, differences between the flat and hierarchical diagnosis models remained very small within each stratum. Flat fusion was marginally better in the ≤ 1.25 mm group, whereas hierarchical fusion showed only negligible advantages in the 1.25–2.5 mm and > 2.5 mm groups. The more noticeable pattern was variation in absolute performance between slice-thickness groups rather than within-group differences between fusion strategies. Overall, these results do not suggest a meaningful

Table 4.19: Subgroup analysis for diagnosis by slice thickness bin. Values are reported as mean $\pm$ standard deviation across seeds. Positive deltas indicate better performance for hierarchical fusion.

Slice thickness bin	N	AUROC			AUPRC		
		Flat	Hier.	Δ	Flat	Hier.	Δ
≤ 1.25 mm	226	0.7429 ± 0.0026	0.7385 ± 0.0038	-0.0043	0.7891 ± 0.0027	0.7854 ± 0.0044	-0.0037
1.25–2.5 mm	351	0.7126 ± 0.0120	0.7159 ± 0.0133	0.0033	0.4315 ± 0.0065	0.4304 ± 0.0055	-0.0012
> 2.5 mm	112	0.8846 ± 0.0078	0.8858 ± 0.0074	0.0012	0.8138 ± 0.0086	0.8144 ± 0.0092	0.0006

slice-thickness-dependent difference in the relative performance of the two fusion strategies.

Overall, the subgroup analysis did not suggest a meaningful subgroup-specific advantage for either flat or hierarchical fusion in diagnosis.

Chapter 5

Discussion

The single-modality baselines provide key context for interpreting the multimodal results. EHR-only models outperformed the individual CT-only models on both tasks, while CT-only stacking produced only modest gains. However, this should be interpreted cautiously. The weaker CT performance does not necessarily mean that imaging contained less useful information overall, because it may also reflect limitations of the individual CT models. In particular, although the 3D model might have been expected to perform strongly given its access to volumetric context, it was the weakest of the CT architectures. One possible explanation is that, unlike the 2D and 2.5D models, the 3D branch was not initialised with pretrained weights, which may have constrained optimisation in the context of a relatively small dataset, although this was not directly evaluated in the present study. The baseline results therefore suggest that, in this implementation, EHR provided the stronger standalone branch, while CT contributed more modest and potentially complementary information.

For diagnosis, the main finding is that multimodal integration mattered more than the precise stage of fusion. The strongest flat and hierarchical models performed almost identically across discrimination and calibration metrics. While SHAP analysis of the strongest XGB-EHR models suggested greater reliance on EHR, a secondary matched LR-EHR comparison showed that the same near-equivalence remained even when CT and EHR contributions were more closely balanced. This suggests that the limited difference between flat and hierarchical fusion for diagnosis was not solely an artefact of EHR dominance, but a more general result of the task.

The prognosis results point to a different pattern. Unlike diagnosis, prognosis showed a clearer separation between fusion strategies, with the strongest hierarchical model outperforming the best flat model on both AUPRC and AUROC, although calibration differences were more mixed. This suggests that, for 12-month mortality prediction, the stage of EHR integration may have been more consequential than it was for diagnosis. Nevertheless, the strongest overall prognosis performance was achieved by the EHR-only XGB

baseline, so the hierarchical advantage should be interpreted within the multimodal setting rather than as evidence that adding CT improved on the best single-modality model. SHAP analysis indicated that, although the EHR branch contributed more strongly than the CT branch, CT-derived information was still incorporated into the final prediction. Taken together, these findings suggest that the prognostic advantage of hierarchical fusion may reflect how CT information was organised before final multimodal integration. This raises the possibility that hierarchical fusion may prove more informative for prognosis when combined with a stronger CT branch capable of contributing greater incremental signal.

The operating-point analysis supports this task-dependent interpretation. For diagnosis, the hierarchical model showed only a small practical advantage at matched high-sensitivity thresholds, suggesting that any benefit beyond overall discrimination was marginal. By contrast, for prognosis, the hierarchical model remained better balanced under high-sensitivity operating points, maintaining higher specificity and PPV than the corresponding flat model. This strengthens the case that, within the multimodal prognosis setting, hierarchical fusion was not only stronger on discrimination metrics but also potentially more useful at clinically relevant threshold-based operating points. However, calibration for prognosis was mixed. Although the two models had the same Brier score, the flat model had a lower ECE. The hierarchical prognosis advantage should therefore be framed as an improvement in discrimination and threshold behaviour rather than as a uniformly superior model on every evaluation dimension.

The subgroup analysis adds further nuance. For diagnosis, the relative performance of flat and hierarchical fusion remained broadly stable across age, vendor, and slice-thickness strata, with only negligible differences between strategies. This supports the interpretation that the near-equivalence of the two fusion methods for diagnosis was not confined to one subgroup. At the same time, prognosis subgroup analysis could not be interpreted reliably because none of the subgroup bins met the predefined stability thresholds. This reflects an important limitation of the study, alongside the relatively small prognosis test set, which likely contributed to instability in calibration curves and subgroup estimates. Accordingly, the stronger prognosis performance of hierarchical fusion should be interpreted as promising but still requiring cautious validation on larger cohorts. In addition, because CT processing proceeded sequentially under storage and computational constraints, the final cohort was not randomly sampled from all eligible scans, so the results are best interpreted as internally valid comparisons rather than as directly generalisable estimates for the full eligible INSPECT population.

Overall, the findings suggest that the importance of fusion timing may depend on the prediction task rather than there being a single best strategy across settings. For diagnosis, combining CT and EHR appears beneficial, but the exact stage of EHR integration

has limited practical importance, so the additional complexity of hierarchical fusion is difficult to justify over a simpler flat approach. For prognosis, hierarchical fusion showed a clearer advantage over flat fusion within the multimodal setting, suggesting that longer-horizon outcome prediction may be more sensitive to how CT information is organised before final multimodal integration. Although the strongest overall prognosis performance was achieved by the EHR-only baseline, these results indicate that preserving internal structure within the CT branch may still be more consequential for prognosis than for diagnosis.

Chapter 6

Conclusion

This dissertation investigated multimodal prediction using CT and EHR data for diagnosis and prognosis, with particular emphasis on whether the stage at which EHR is introduced affects model performance. Across the baseline experiments, EHR provided the strongest standalone predictive signal, while CT contributed weaker but still informative signal. For diagnosis, multimodal fusion improved substantially over the CT-only baselines, supporting the value of combining image-derived and structured clinical information for this task.

The main finding of the dissertation is that the value of fusion architecture was task-dependent. For diagnosis, flat and hierarchical fusion achieved near-identical performance, indicating that the principal gain came from multimodal integration itself rather than from the precise architectural stage of EHR incorporation. For prognosis, hierarchical fusion showed a clearer advantage over flat fusion within the multimodal setting, suggesting that longer-horizon outcome prediction may be more sensitive to how CT information is organised before final multimodal integration. However, the strongest overall prognosis performance was achieved by the EHR-only XGB baseline, so this result should be interpreted as evidence about the effect of fusion strategy within CT-augmented multimodal models rather than as evidence that multimodal fusion outperformed the best single-modality model. Overall, these findings show that increased fusion complexity is not uniformly advantageous and should be justified by task-specific performance gain rather than assumed a priori.

6.1 Future Work

Several directions could extend this work. First, future studies should evaluate prognosis on larger cohorts, as the present prognosis analyses were constrained by limited sample size, particularly in subgroup evaluation. Second, the CT branch should be strengthened, since the weaker CT-only performance in this study may partly reflect limitations of the

modelling implementation rather than the available imaging information itself. This applies particularly to the 3D branch, which performed more weakly than expected and may benefit from stronger volumetric encoders, improved training strategies, or pretraining better matched to the downstream task. Future work should test whether stronger CT representations can provide genuine incremental prognostic value beyond the strong EHR-only baseline observed here. Third, the fusion question could be revisited using architectures with more explicit cross-modal interaction. This was beyond the scope of the present study but would help determine whether the limited diagnosis difference observed here is specific to the fusion forms examined.

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Chapter A

Appendix

A.1 Additional CT Preprocessing Details

Study description filtering. Study descriptions were converted to uppercase and retained if they contained any of the following terms: PULMONARY EMBOLISM, CTPA, CT ANGIO, or CHEST ANGIO. In addition, the regular expression `\bPE\b` was used to match PE as a standalone word.

A.1.1 Heuristic Slice Score for the 2D Pipeline

To identify informative slices for the 2D model, each slice was assigned a heuristic score based on simple intensity- and position-based features intended to reflect the likelihood that the slice contained relevant thoracic anatomy. This approach was used instead of explicit anatomical segmentation as a simpler and more computationally efficient preprocessing strategy.

The slice score was defined using the six components shown in Table A.1. These components were chosen to favour slices containing substantial body area, central soft tissue, and vascular structures, while down-weighting slices dominated by background air, peripheral anatomy, or extreme superior and inferior positions within the scan.

Component	Weight
Body fraction	0.20
Lung-band preference	0.25
Central soft tissue	0.20
Central vessel density	0.20
Edge gradient	0.10
Position prior	0.05

Table A.1: Components of the heuristic slice score used for 2D slice selection.

Body fraction measured the proportion of non-air voxels in the slice, favouring slices

containing a substantial portion of the patient body. Lung-band preference favoured slices with a plausible proportion of lung-region voxels, helping to prioritise thoracic slices while penalising slices outside the main chest region. Central soft tissue and central vessel density favoured slices containing mediastinal soft tissue and contrast-enhanced vascular structures near the centre of the image. Edge gradient captured local intensity variation within the vascular window as a simple measure of structural detail. Finally, a position prior assigned slightly greater weight to slices near the middle of the scan, reflecting the expectation that informative thoracic anatomy was more likely to appear away from the extreme superior and inferior ends.

A.1.2 Three-Window Input Representation for the 2D Model

The **ResNet** backbone expects three input channels. For the 2D model, each selected CT slice was therefore converted into a three-channel image using three intensity windows. The first channel used a soft-tissue window centred at 50 HU with width 350 HU, the second used a vascular window centred at 250 HU with width 600 HU, and the third used a lung window centred at -600 HU with width 1500 HU. Together, these windows provided complementary views of central soft tissue, contrast-enhanced vessels, and broader thoracic structure while remaining compatible with the three-channel CNN input.

A.2 Model-Specific Training Details

Table A.2: Shared CT training settings.

Setting	Value
Loss	Weighted cross-entropy
Class weights	Inverse-frequency, normalised to sum to 2
Optimiser	AdamW
Maximum epochs	20
Early stopping metric	Validation AUPRC
Early stopping patience	4 epochs
Learning-rate scheduler	ReduceLROnPlateau
LR reduction factor	0.5
LR scheduler patience	1 epoch
Minimum learning rate	10^{-6}
Validation split	Internal 80/20 split of training partition
Distributed training	DDP across 3 GPUs
Threshold tuning	Implemented on validation predictions for binary decision outputs, but not used in the reported test-set AUROC/AUPRC/Sensitivity/Specificity results

Table A.3: Model-specific CT optimisation settings. Effective batch size is computed as per-device batch size $\times$ accumulation steps $\times$ 3 GPUs.

Model	Per-device batch	Accum. steps	Effective batch	Backbone LR	Head LR	Weight decay
2D	16	1	48	3×10^{-5}	1.5×10^{-4}	2×10^{-4}
2.5D	8	2	48	1×10^{-5}	1×10^{-4}	6×10^{-4}
3D	6	3	54	–	7×10^{-5}	2×10^{-4}

A.2.1 Random Seeds

To assess stability with respect to random initialisation and fold assignment, experiments were repeated across five random seeds: 0, 1, 42, 123, and 2026. Seed-specific outputs were stored separately and aggregated in the experiment summary tables. For each model and task, summary statistics were therefore computed over five independent seeded runs.

Table A.4: Structured EHR model training settings.

Model	Settings
Logistic regression	L_2 -regularised logistic regression with <code>liblinear</code> solver, $C = 0.1$, <code>max_iter</code> =2000, <code>class_weight</code> =balanced. Numeric features were median-imputed and standardised; binary features were passed through unchanged.
XGBoost	<code>n_estimators</code> =2000, learning rate = 0.03, max depth = 4, subsample = 0.8, colsample by tree = 0.8, <code>min_child_weight</code> =1, $\lambda = 1.0$, $\alpha = 0.0$, $\gamma = 0.0$, objective = binary logistic, evaluation metric = AUPRC, early stopping after 50 rounds on an internal 20% validation split. Missing numerical features are handled by native model logic.

Table A.5: Stacked and ensemble model settings.

Component	Settings
OOF protocol	5-fold stratified out-of-fold stacking. Base-model training predictions were generated only on held-out folds. Test-set meta-features were constructed by averaging fold-specific base-model predictions across the five trained fold models.
CT mean ensemble	Unweighted mean of 2D, 2.5D, and 3D predicted probabilities.
Logistic-regression meta-learner	Logistic regression fitted on OOF base-model probabilities, with <code>max_iter</code> =1000.
XGBoost meta-learner	XGBoost fitted on OOF base-model probabilities, with <code>n_estimators</code> =200, <code>max_depth</code> =3, learning rate = 0.05, subsample = 0.8, colsample by tree = 0.8, <code>eval_metric</code> =logloss, <code>use_label_encoder</code> =False, and <code>random_state</code> =42.

A.3 Additional Evaluation and Interpretability Details

Table A.6: SHAP summary computation for stacked models.

Item	Details
Features explained	Meta-model input features, i.e. predicted probabilities from component models
Supported models	Stacked models with logistic-regression or XGBoost meta-learners
Explainer	<code>shap.LinearExplainer</code> for logistic regression; <code>shap.TreeExplainer</code> for XGBoost
Background set	Training meta-feature matrix, capped at 2,000 samples
Explained set	Test meta-feature matrix, capped at 2,000 samples
Explained output	Positive-class prediction from the meta-learner
Reported output	Mean absolute SHAP value per meta-feature on the explained set, ranked from largest to smallest

Table A.7: Subgroup stability criteria.

Item	Details
Minimum total subgroup size	100 scans
Minimum positive cases	30
Minimum negative cases	30
Stable subgroup definition	A subgroup was considered stable only if all three thresholds were met within a random-seed run
Across-seed summary	<code>stable_repeats</code> counts how many random-seed runs met the rule; <code>all_repeats_stable</code> indicates that all seeded runs met it